Fuzzy PI Control for bittide: Nonlinear Analysis and Transient Performance

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Abstract

Draft status. The abstract is pending. This working manuscript integrates the Introduction, Background, expanded controller model, reference-model derivations and Evaluation; Discussion and Conclusion remain to be written. Author, conference and CCS metadata require final confirmation.

CCS Concepts

• **Computing methodologies**;

Keywords

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1 Introduction

Distributed cyber-physical computation requires nodes running on independent clocks to coordinate their computations and communications. For applications that depend on reproducible execution, the relationships between communicated events must remain well defined as physical timing varies. This creates a fundamental implementation challenge: a coherent logical execution must be realized using oscillators, communication links and storage resources whose physical behaviour is neither identical nor unlimited.

Logical synchrony provides a framework for addressing this challenge without distributing a global physical clock. In bittide, continuous frame exchange and elastic receiving buffers connect the physical operation of neighbouring nodes to their logical execution. Each node adjusts its oscillator frequency using locally observed buffer occupancies. These buffers consequently serve two roles: they provide the feedback signal for synchronization and absorb the imbalance between incoming frames and local consumption. The storage available to accommodate that imbalance is finite [**?**].

The resulting control problem concerns the entire trajectory towards an operating point. Initial oscillator differences and changes in frame-delivery time can consume buffer headroom before the network settles or while it responds to a disturbance. A small eventual frequency disagreement does not by itself establish that the intervening buffer trajectory remains admissible. The objective is therefore to use local occupancy feedback to reduce frequency disagreement and control deviations from compatible occupancy targets, while respecting finite buffer capacity and oscillator-correction limits. This formulation makes transient buffer excursion, convergence and control activity joint criteria for evaluating a controller.

Existing bittide research establishes important foundations for this problem. PI performance analysis [**?**] relates closed-loop behaviour to controller gains and network resistance distances. Fluid occupancy analysis [**?**] characterizes steady buffer levels, and reframing methods [**?**] address buffer centering. Our earlier study, *Stabilizing the bittide by Tuning Controllers Under Topology and Latency Constraints*, examined PI gain selection under topology and fixed-delay constraints. Building on these results, the question addressed here is how a local nonlinear Fuzzy PI law can reshape transient resource use under comparable scales, memory and update conventions. In particular, the connection between the nonlinear rule surface, a provable buffer-excursion benefit and behaviour under delivery perturbations requires explicit analysis and controlled comparison.

We propose a decentralized occupancy-based Fuzzy PI controller for bittide. Each node computes an oscillator-correction increment from its receiving-buffer error and the update-to-update change in that error. In the central input region studied here, the proposed surface strengthens its calibrated linear counterpart for same-sign inputs and attenuates its magnitude for opposing-sign inputs. This direction-dependent action provides flexibility at fixed calibrated PI gain coordinates. It motivates investigating whether the nonlinear correction can reduce transient buffer demands while retaining convergence, and determining how its effects on frequency disagreement and control activity depend on the operating conditions.

This paper presents the complete controller, establishes its continuous-clock realization and combines nonlinear analysis with evaluation to investigate that question. We derive the exact central-region map and its scale correspondence with native PI, and verify implementation equivalence when the nonlinear map is replaced by its calibrated linear counterpart. For a synchronous, zero-delay fluid reference, we establish an invariant input region and a common Lyapunov certificate. An outward interval enclosure of the nonlinear trajectory, followed by a contracting infinite-tail bound, provides an all-time buffer-excursion guarantee. For one prescribed three-node initialization at matched gain coordinates, the Fuzzy excursion is certified below 17.315 fluid frames, whereas PI attains an exact peak of approximately 23.403701 frames. This establishes a strict buffer-headroom benefit for that reference and initialization.

We separately evaluate the controllers in the continuous-phase Callisto model, with quantized buffer events, local-clock polling and control-execution delay. Controller updates change the slope of continuous phase trajectories; they do not replace the plant with a discrete firing-count model. Shared frame-delivery stress profiles include a latency burst after nominal centering and bounded random-latency traces. Over the declared measurement window

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beginning at burst onset, Fuzzy PI has lower integrated frequency-disagreement amplitude and command total variation, alongside a higher peak buffer excursion. In both random-latency regimes, the matched moderate PI baseline achieves lower disagreement RMS over the declared disturbance windows. These experiments characterize the performance tradeoffs accompanying nonlinear response shaping and distinguish the reference-model guarantee from its practical applicability under the declared execution assumptions.

2 Background

2.1 Logical Synchrony and the bittide Mechanism

Logical synchrony describes relationships between distributed events without requiring a global physical clock. In bittide, neighbouring nodes exchange ordered frames, and elastic receiving buffers accommodate differences in their physical progress. The association between sending and receiving events provides the basis for a logical execution whose dependencies can be reasoned about independently of exact wall-clock timing. Equal physical phases are therefore not required for logical synchrony. Frequency agreement alone, however, does not establish the required event relationships: frame ordering, initialization and continued availability of the corresponding frames also matter. These distinctions follow the logical-synchrony and multiclock-network framework developed by Lall et al. [**?**].

The control problem supports this realization by regulating physical progress while retaining the prescribed frame relationships. Each node observes its own receiving buffers and adjusts its oscillator. Buffers consequently provide both the local feedback signal and finite storage for a temporary imbalance between delivery and consumption.

2.2 Continuous Clocks and Receiving-Buffer Dynamics

We consider a network of n nodes. Node i has continuous oscillator phase $\theta_i(t)$, measured in local cycles, and physical frequency

$$\dot{\theta}_i(t) = \omega_i(t) = \omega_i^u + u_i(t), \tag{1}$$

where ω_i^u is its constant intrinsic frequency and $u_i(t)$ is the applied frequency correction. The correction can be signed; admissible physical operation requires $\omega_i(t) > 0$. In the current Callisto model, corrections are held between local actuation events. Phase remains continuous when a correction changes, although its slope can change. Local sampling and integer frame events are embedded in this continuous-time plant.

For a receiving edge $e = (j \to i)$, let $\beta_e(t)$ be buffer occupancy in frames and C_e its capacity. During normal operation without progress blocking,

$$\begin{aligned} \beta_e(t) &= \beta_e(0) + A_{ji}^{\mathrm{arr}}(t) - N_i(t), \\ N_i(t) &= \lfloor \theta_i(t) \rfloor - \lfloor \theta_i(0) \rfloor. \end{aligned} \tag{2}$$

Here $A_{ji}^{\mathrm{arr}}(t)$ counts arrivals during $(0, t]$, including frames already in flight at initialization, and $N_i(t)$ counts local consumption. This conservation relation explains why both oscillator mismatch and delivery changes affect occupancy. Section **??** specifies the local schedules, initialization and ordered arrival model used in our realization.

Fluid models replace integer frame counts by continuous phase increments. With constant physical path delay τ_{ji}, the corresponding occupancy satisfies, between command discontinuities,

$$\dot{\widetilde{\beta}}_{ji}(t) = \omega_j(t - \tau_{ji}) - \omega_i(t), \tag{3}$$

where $\widetilde{\beta}_{ji}$ denotes fluid occupancy. Thus occupancy integrates the imbalance between delayed source progress and local consumption. The fluid approximation facilitates analysis; its assumptions must be distinguished from those of integer-frame execution. This separation is also central to existing PI performance analysis [**?**] and buffer-occupancy modeling [**?**].

2.3 Safety, Syntony and Buffer Centering

Safety is a property of the complete trajectory. For the normal operating regime studied here, we seek

$$0 < \beta_e(t) < C_e, \qquad \omega_i(t) > 0, \qquad |u_i(t)| \le u_{\max}, \tag{4}$$

for every edge, node and time, where $u_{\max} > 0$ is the correction limit. The available lower and upper buffer reserves are $\beta_e(t)$ and $C_e - \beta_e(t)$, respectively. A convergent trajectory can still exhaust either reserve before reaching its limit. Avoiding such exhaustion through control is distinct from invoking a protection mechanism that blocks logical progress.

Define the physical frequency spread as

$$\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t). \tag{5}$$

For nominal fluid operation with fixed conditions, we seek $\Delta_\omega(t) \to 0$, together with $\beta_e(t) \to \beta_e^*$ for compatible occupancy targets β_e^*. Frequency agreement is the nominal syntonization objective; it does not imply zero phase differences or tracking an externally prescribed frequency. Without an external reference, the common limiting frequency ω_∞ is determined by the closed loop. Likewise, arbitrary per-edge targets cannot be assumed compatible with a fixed marking, meaning the frame-count offsets determined by initialization, and with the selected initial state.

If a transient frequency-spread tolerance $\Delta_{\mathrm{safe}} > 0$ is specified, then $\Delta_\omega(t) < \Delta_{\mathrm{safe}}$ is an additional engineering constraint. It is not the definition of logical synchrony. Under persistent delivery perturbations, we assess bounded deviations, residual disagreement and buffer reserves rather than assume exact disturbance-free centering.

2.4 Delivery Perturbations and Local Feedback

A latency increase affects the delivery process before a receiver necessarily observes it through its next control measurement. If arrivals pause while consumption continues, the conservation relation shows that the receiving buffer decreases; subsequent release of accumulated frames can instead increase occupancy. The disturbance therefore acts through frame availability and accumulated count imbalance. A controller cannot recreate an undelivered frame, and its response is constrained by measurement and actuation timing. Our experiments use declared order-preserving delivery traces

so that controller comparisons share the same external stress definition. The exact arrival construction is given in Section **??**.

2.5 Existing Control Foundations and the Present Study

Existing work analyzes occupancy-feedback PI performance through controller gains and graph resistance distances, derives limiting fluid occupancies, and develops reframing methods that separate frequency regulation from occupancy positioning. These are complementary foundations: PI performance [**?**], steady buffer occupancy [**?**], and buffer centering [**?**]. Their conclusions apply under their respective modeling assumptions; in particular, continuous-time gain conditions cannot be transferred unchanged to sampled execution with delay.

This paper builds on those foundations by proposing a local incremental Fuzzy PI law and analyzing how its nonlinear increments reshape the closed-loop response at explicitly calibrated PI scales. Section **??** defines the complete controller and establishes that calibration. Section **??** derives sufficient safety and convergence conditions for a declared synchronous, zero-delay fluid reference, followed by an all-time buffer-excursion comparison for one prescribed initialization. Evaluation in continuous-phase Callisto separately examines delivery perturbations under the actual local update and frame-count conventions.

3 Proposed Fuzzy PI Control for bittide and Nonlinear Analysis

We propose a decentralized, occupancy-based incremental Fuzzy PI controller for bittide. Each node combines receiving-buffer error with its update-to-update change to compute an oscillator-correction increment. This section specifies the complete controller and its continuous-clock interface, derives an implementation-consistent PI calibration, and establishes the nonlinear properties used in Section **??**. The physical plant, integer frame counts and sampled controller state are kept distinct throughout the derivation.

3.1 Continuous-clock realization of the proposed controller

The plant is the current continuous-phase Callisto model within the bittide framework [**?**]. For constant intrinsic oscillator frequency ω_i^u and applied correction $u_i(t)$,

$$\begin{aligned}\dot{\theta}_i(t) &= \omega_i^u + u_i(t),\\ u_i(t) &= u_{i,k}, \qquad t \in [t_{i,k}^a, t_{i,k+1}^a),\\ \theta_i(t) &= \theta_i(t_{i,k}^a) + (\omega_i^u + u_{i,k})(t - t_{i,k}^a).\end{aligned} \tag{6}$$

The superscript a denotes actuation, and k indexes that node's updates. Normal executions require $\omega_i^u + u_i(t) > 0$, so the phase has a monotone inverse. Physical phase is continuous at an update; its slope can change. Requested correction $q_{i,k}$ is computed from the measured buffers and applied as $u_{i,k} = \mathrm{sat}_{u_{\max}}(q_{i,k})$, with $\mathrm{sat}_b(z) = \min\{b, \max\{-b, z\}\}$ and $u_{\max} > 0$. The proposed controller has jump updates inside a continuous physical plant; it is not implemented as an ordinary differential equation for its accumulator.

Let $p > 0$ be the polling period and $d_{\mathrm{ctrl}} \geq 0$ the execution delay, both measured in local-clock phase units. Measurement time $t_{i,k}^m$ and actuation time $t_{i,k}^a$ satisfy

$$\begin{aligned}\theta_i(t_{i,k}^m) &= \theta_i(0) + k\,p,\\ \theta_i(t_{i,k}^a) &= \theta_i(0) + k\,p + d_{\mathrm{ctrl}},\\ \int_{t_{i,k}^m}^{t_{i,k}^a} (\omega_i^u + u_i(s))\,ds &= d_{\mathrm{ctrl}}.\end{aligned} \tag{7}$$

The product $k\,p$ in these scheduling equations means the integer index k multiplied by the poll period p; it is not the proportional gain k_p. With zero initial correction, $t_{i,0}^m = 0$ and $t_{i,0}^a = d_{\mathrm{ctrl}}/\omega_i^u$. Subsequent actuation intervals equal $p/(\omega_i^u + u_{i,k})$. The delay is not automatically a constant physical time, and its age must not be computed using the new command that is applied only at the end of that age interval.

In normal operation, each integer source-phase crossing sends a frame. With the progress guards open, frame conservation gives

$$\beta_{j\to i}(t) = \beta_{j\to i}(0) + A_{ji}^{\mathrm{arr}}(t) - N_i(t), \tag{8}$$

where $A_{ji}^{\mathrm{arr}}(t)$ counts arrivals during $(0, t]$, including initial in-flight frames, and $N_i(t) = \lfloor\theta_i(t)\rfloor - \lfloor\theta_i(0)\rfloor$ counts local consumption. For a constant physical delay $\tau_{ji} \geq 0$,

$$A_{ji}^{\mathrm{arr}}(t) = \lfloor\theta_j(t - \tau_{ji})\rfloor - \lfloor\theta_j(-\tau_{ji})\rfloor. \tag{9}$$

Thus arrival counts incorporate the declared negative-time history.

For constant physical path delay τ_{ji}, receiving occupancy is

$$\beta_{j\to i}(t) = \lambda_{ji}^{\mathrm{UGN}} + \lfloor\theta_j(t - \tau_{ji})\rfloor - \lfloor\theta_i(t)\rfloor. \tag{10}$$

Here $\lambda_{ji}^{\mathrm{UGN}}$ is the fixed integer offset in the absolute phase-count convention. It is determined by occupancy and prehistory:

$$\lambda_{ji}^{\mathrm{UGN}} = \beta_{j\to i}(0) - \lfloor\theta_j(-\tau_{ji})\rfloor + \lfloor\theta_i(0)\rfloor. \tag{11}$$

The relative-count logical marking is $\lambda_{ji}^{\mathrm{LSN}} = \lambda_{ji}^{\mathrm{UGN}} + \lfloor\theta_j(0)\rfloor - \lfloor\theta_i(0)\rfloor$; the two coincide when the initial phase counts are equal.

For changing delivery times, let $s_{j,m} = \theta_j^{-1}(m)$ be the send time of source tick m, and $a_{ji}(s)$ its causal ordered arrival map. For the desired send-time delay profile $\tau_{ji}(s)$, the declared running-maximum envelope is

$$a_{ji}(s) = \sup_{r\leq s}\{r + \tau_{ji}(r)\},$$

including the constant-delay negative-time history. Define the delivered tick frontier

$$C_{ji}^{\mathrm{arr}}(t) = \max\{m \in \mathbb{Z} : a_{ji}(s_{j,m}) \leq t\}, \quad \beta_{j\to i}(t) = \lambda_{ji}^{\mathrm{UGN}} + C_{ji}^{\mathrm{arr}}(t) - \lfloor\theta_i(t)\rfloor. \tag{12}$$

The negative-time history specifies initial in-flight frames. The declared running-maximum arrival envelope preserves order and can release a batch after a desired-delay decrease; it imposes no finite serialization rate. At an upward-delay discontinuity, a supremum of continuous source times need not itself have arrived. Counting delivered tick indices avoids silently including such a boundary frame.

If a protection guard blocks logical progress, the runtime can retain a separate continuous data phase ψ_i with $\dot{\psi}_i = g_i(t)\dot{\theta}_i$, where $g_i \in \{0, 1\}$ is the progress gate. Buffer counts then use ψ_i, whereas

scheduling still uses θ_i. In the ungated normal regime, with matching initial phases, data phase and physical phase agree. Guard operation does not stop the oscillator.

Only for the reference proof do we replace integer occupancy by a fluid counterpart. At constant delay, define $\widetilde{\beta}_{ji}(t) = \widetilde{\lambda}_{ji} + \theta_j(t - \tau_{ji}) - \theta_i(t)$, where $\widetilde{\lambda}_{ji}$ is an additive offset chosen for the specified fluid initialization. Between command discontinuities,

$$\dot{\widetilde{\beta}}_{ji}(t) = \omega_j^u + u_j(t - \tau_{ji}) - \omega_i^u - u_i(t). \tag{13}$$

This derivative belongs to the fluid counterpart, not the actual integer occupancy. Replacing floor counts by phases does not automatically preserve an arbitrary initialization; the compatible zero-delay marking used in Section **??** is specified there.

3.2 Notation and the measured error

Node i has incoming-neighbour set $\mathcal{N}_i^-$ and $d_i = |\mathcal{N}_i^-| > 0$ incoming channels. All incoming sums below range over $j \in \mathcal{N}_i^-$. This section assumes fixed active incoming sets: the native link-status mask is one on all configured inputs. The Fuzzy percentage wrapper averages its configured inputs, so changing link masks would require an explicit corresponding measurement policy. At the kth local measurement, $\beta_{ji,k}$ is the measured occupancy of channel $j \to i$, and β_{ji}^* is its fixed target. Use

$$E_{i,k} = \sum_{j \in \mathcal{N}_i^-} (\beta_{ji,k} - \beta_{ji}^*). \tag{14}$$

The sign is actual minus target. If incoming buffers are too full, a positive frequency correction increases local consumption. This explains the positive PI gains with this error convention. The physical frequency is $\omega_i = \omega_i^u + u_i$, where ω_i^u is the constant intrinsic frequency; u_i is a correction, not the entire frequency.

Symbol	Meaning
$E_{i,k}$	Sum error in frames
$e_{i,k}$	Target-minus-measured error in percentage points
$I_{i,k}$	Native PI sum of sampled frame errors
$A_{i,k}$	Fuzzy accumulator: total requested correction
$q_{i,k}, u_{i,k}$	Requested and applied frequency corrections
h	Fixed native PI integration scale
$X_{i,k}, Y_{i,k}$	Scaled Fuzzy error and sample-difference inputs
$F_0, \mathcal{R}_F$	Linear counterpart and residual $F - F_0$

The event index k is local to a node. The normalization factor γ_i converts a frame-sum error into percentage points. The maximum correction $u_{\max} > 0$ is the actuator limit; q and u remain distinct if clipping acts. Two nodes need not measure at the same physical times. The subscript ji abbreviates edge $j \to i$. Suppressing i below is a notational convenience, not a synchrony assumption; in that notation $\gamma = \gamma_i$.

3.2.1 Deriving the percentage-to-token relation. For common incoming capacity C, define

$$\bar{p}_{i,k} = \frac{1}{d_i} \sum_j \frac{100\beta_{ji,k}}{C}, \qquad r_i = \frac{1}{d_i} \sum_j \frac{100\beta_{ji}^*}{C}.$$

Then subtraction and factoring give

$$e_{i,k} = r_i - \bar{p}_{i,k} \tag{15}$$

$$= -\frac{100}{d_i C} \sum_j (\beta_{ji,k} - \beta_{ji}^*) = -\gamma_i E_{i,k}, \qquad \gamma_i = \frac{100}{d_i C}. \tag{16}$$

For midpoint targets this is the original absolute-percentage input with $r_i = 50$. For other targets the runtime can supply a centered percentage $r_i + d_i^{-1} \sum_j 100(\beta_{ji,k} - \beta_{ji}^*)/C$; subtracting it from r_i yields the same $e_{i,k}$. A mismatched setpoint instead adds a constant bias and invalidates the scalar relation used for calibration.

With unequal capacities and a consistently matched target reference, the percentage average gives

$$e_{i,k} = -\frac{100}{d_i} \sum_j \frac{\beta_{ji,k} - \beta_{ji}^*}{C_{ji}}.$$

This is a weighted error and generally cannot be written as one normalization constant times the unweighted native sum. Matching such a controller requires a consistent measurement weighting or a different baseline.

3.3 Native PI: update order, units and the first call

The occupancy-feedback PI baseline is grounded in the bittide analysis of [**?**]. The executed native update is an integrate-then-output recursion:

$$I_k = I_{k-1} + E_k, \qquad q_k^{\mathrm{PI}} = k_p E_k + \kappa_I I_k, \qquad \kappa_I = k_i h. \tag{17}$$

The sum I_k contains no physical time multiplier; h is incorporated in κ_I. In the present runtime,

$$h = \frac{p}{\omega_{\mathrm{base}}}. \tag{18}$$

Here p is the local polling period and ω_{base} is the fixed implementation reference frequency. The scale h stays fixed during the configured run. Local polling intervals measured in physical time can vary as the oscillator changes; replacing h by those intervals would implement a different PI law.

Subtract the previous requested output:

$$q_k^{\mathrm{PI}} - q_{k-1}^{\mathrm{PI}} = k_p(E_k - E_{k-1}) + \kappa_I(I_k - I_{k-1}) \tag{19}$$

$$= k_p(E_k - E_{k-1}) + \kappa_I E_k. \tag{20}$$

This is the correct increment to reproduce. It contains the present E_k in the integral term because integration precedes output.

At zero initial sum, the first call returns

$$q_0^{\mathrm{PI}} = (k_p + \kappa_I)E_0. \tag{21}$$

No physical prehistory measurement is needed to express this algebra: one may use the fictitious previous error $E_{-1} = 0$ and previous requested output $q_{-1} = 0$. That bookkeeping choice is separate from deriving a plant identity such as $x_k - x_{k-1} = Hw_{k-1}$, which requires a genuine elapsed plant interval.

Table 1: Rule consequents of the proposed bittide Fuzzy PI controller.

$X\backslash Y$	NL	NS	ZE	PS	PL
NL	NL	NL	NL	NS	ZE
NS	NL	NL	NS	ZE	PS
ZE	NL	NS	ZE	PS	PL
PS	NS	ZE	PS	PL	PL
PL	ZE	PS	PL	PL	PL

3.4 The proposed Fuzzy PI controller

The proposed local incremental control law combines the present error and its sample difference as follows:

$$X_k = K_e e_k, \qquad Y_k = K_{\Delta e}(e_k - e_{k-1}), \tag{22}$$

$$A_k = A_{k-1} + K_u F(X_k, Y_k), \qquad q_k^{\mathrm{F}} = A_k. \tag{23}$$

The gain signs used here are $K_e, K_{\Delta e} > 0$ and $K_u < 0$. Thus high occupancy produces negative e, and a negative surface output produces a positive correction increment. The quantity $e_k - e_{k-1}$ is not divided by elapsed time; calling it a time derivative would obscure its sampling dependence.

3.4.1 Memberships and rule weights. The labels NL, NS, ZE, PS and PL mean negative large, negative small, zero, positive small and positive large, respectively. Each membership $\varphi_a(z) \in [0, 1]$ is an activation degree; a singleton is a numerical rule consequent. Let $[z]_+ = \max\{z, 0\}$ and $\mathrm{clip}_{[0,1]}\, z$ denote projection onto $[0, 1]$. The exact memberships can be written

$$\begin{aligned} &\varphi_{\mathrm{NL}}(z) = \mathrm{clip}_{[0,1]}\,\frac{-z-5}{5}, && \varphi_{\mathrm{NS}}(z) = [1 - |z+5|/5]_+, \\ &\varphi_{\mathrm{ZE}}(z) = [1 - |z|/5]_+, && \varphi_{\mathrm{PS}}(z) = [1 - |z-5|/5]_+, \\ &\varphi_{\mathrm{PL}}(z) = \mathrm{clip}_{[0,1]}\,\frac{z-5}{5}. \end{aligned} \tag{24}$$

Table **??** gives all 25 consequent rules. Order the labels as NL, NS, ZE, PS and PL, with label indices $a, b \in \{1, \ldots, 5\}$. With singleton values $(s_1, \ldots, s_5) = (-5, -2.5, 0, 2.5, 5)$ and $\ell(a, b) = \min\{5, \max\{1, a+b-3\}\}$, compute

$$w_{ab} = \min\{\varphi_a(X), \varphi_b(Y)\}, \qquad F(X, Y) = \frac{\sum_{a,b} w_{ab} s_{\ell(a,b)}}{\sum_{a,b} w_{ab}}. \tag{25}$$

The membership partition gives at least one positive input membership on each axis, hence a positive denominator at finite inputs. Individual rules are summed even when they share a consequent. Replacing this accumulation by maximum aggregation changes the map and its derivation.

The weighted-average property gives $|F| \le 5$ and therefore $|A_k - A_{k-1}| \le 5|K_u|$. This is a per-update bound, not a bound on A_k over an arbitrarily long execution. The raw Fuzzy module itself has no final correction clipping. Initialize $A_{i,-1} = 0$ and $e_{i,-1} = 0$. At each measurement, form $e_{i,k}$ and $(X_{i,k}, Y_{i,k})$, evaluate the rule weights and normalized surface, increment $A_{i,k}$, and store $e_{i,k}$ as the previous error. The requested correction is then limited and applied at $t_{i,k}^a$, and held until the next actuation. In the normal-run policy considered here, external clipping retains the raw accumulator and anti-windup is disabled. The controller does not overwrite $A_{i,k}$ with the applied command. The membership functions, rule consequents and gains stay fixed during a run; the correction changes with the measured input pair. The law requires neither an observer nor an FFP mode.

3.5 Exact PI Calibration, Including Initialization

Define the globally linear counterpart

$$F_0(X, Y) = \frac{X + Y}{2}. \tag{26}$$

Replacing F by this map retains the same controller accumulator. Using $e = -\gamma E$ gives

$$A_k^0 - A_{k-1}^0 = \frac{K_u}{2}\{K_e e_k + K_{\Delta e}(e_k - e_{k-1})\} \tag{27}$$

$$= -\frac{K_u \gamma K_e}{2} E_k - \frac{K_u \gamma K_{\Delta e}}{2}(E_k - E_{k-1}). \tag{28}$$

Matching the two independent coefficients in (**??**) yields

$$k_p = -\frac{K_u \gamma K_{\Delta e}}{2}, \qquad \kappa_I = -\frac{K_u \gamma K_e}{2}. \tag{29}$$

Solving for two Fuzzy gains with chosen $K_e > 0$ gives

$$K_u = -\frac{2k_i h}{\gamma K_e}, \qquad K_{\Delta e} = \frac{K_e k_p}{k_i h}. \tag{30}$$

The formulas assume $k_i h > 0$. Pure proportional control is a separate limiting case and should not be inserted into this inverse mapping with a zero denominator.

Proposition 3.1 (Event-level linear-counterpart equivalence). *For any measurement sequence, suppose* (**??**) *holds and initialize*

$$A_{-1}^0 = k_p E_{-1} + \kappa_I I_{-1}, \qquad e_{-1} = -\gamma E_{-1}. \tag{31}$$

Then $A_k^0 = k_p E_k + \kappa_I I_k$ *for every* $k \ge 0$.

Proof. Assume the relation holds before update k. Substitution gives

$$\begin{aligned} A_k^0 &= k_p E_{k-1} + \kappa_I I_{k-1} + k_p(E_k - E_{k-1}) + \kappa_I E_k \\ &= k_p E_k + \kappa_I (I_{k-1} + E_k) = k_p E_k + \kappa_I I_k. \end{aligned}$$

The stated initialization supplies the induction base, including $k = 0$. □

3.5.1 Why zero constructors work for a nonzero first error. Take $I_{-1} = A_{-1}^0 = e_{-1} = 0$, corresponding to $E_{-1} = 0$. The first Fuzzy-reference input is

$$(X_0, Y_0) = (-\gamma K_e E_0, -\gamma K_{\Delta e} E_0),$$

so $A_0^0 = (k_p + \kappa_I) E_0$, exactly as in (**??**). If only Fuzzy is warmed up to $e_{-1} = e_0$ while $A_{-1}^0 = 0$, its difference input vanishes and the output is merely $\kappa_I E_0$. The missing $k_p E_0$ term can create a large apparent controller difference unrelated to fuzzy inference.

3.5.2 A numerical calibration from the present configuration. For $d = 2$, $C = 2048$, $p = 1000$, $\omega_{\text{base}} = 29999/30000$, $k_p = 2 \times 10^{-6}$, $k_i = 5 \times 10^{-11}$ and $K_e = 0.1$,

$$\gamma = \frac{25}{1024}, \qquad h = \frac{30000000}{29999},$$
$$\kappa_I = \frac{3}{59998000}, \qquad K_u = -\frac{768}{18749375},$$
$$K_{\Delta e} = \frac{29999}{7500} \simeq 3.9998666667. \tag{32}$$

For a one-frame first error, the matched PI and F_0 corrections are both $2.0500016667 \times 10^{-6}$. Warming up only the Fuzzy previous error instead gives approximately $5.0001666722 \times 10^{-8}$. The example illustrates an initialization mismatch; it is not a new plant simulation.

3.5.3 Execution-layer conditions. Identical raw outputs give identical applied outputs under a common limiter when raw states are retained and anti-windup is disabled. Tracking or reset requires compatible state changes. For example, after processing E_k, setting

$$A_k^0 = u_{\text{held}}, \qquad I_k = \frac{u_{\text{held}} - k_p E_k}{\kappa_I}, \qquad e_k = -\gamma E_k$$

restores the state relation for the next ordinary update. A branch designed to produce a held output on an immediate extra update requires its own integrate-first adjustment; the above is not an instruction to add an extra call. Limiting the internal reference F_0 to $[-5, 5]$ also changes the controller, even if the external correction limiter is shared.

Under a deterministic causal plant, the same initial history, measured inputs, actuation order, external trace and compatible resets imply closed-loop equality by induction over events. No globally synchronous physical grid is required for this algebraic equivalence. By contrast, matching only labels such as "same gains" or "same random seed" does not establish these conditions.

3.6 Deriving the Exact Central Nonlinear Surface

For $0 < r \le 5$, define

$$\Omega_r = \{(X, Y) : |X| + |Y| \le r\},$$
$$m = \min(|X|, |Y|), \qquad s = |X| + |Y|. \tag{33}$$

The same-sign interaction is

$$J(X, Y) = \begin{cases} \operatorname{sgn}(X)m, & XY > 0, \\ 0, & XY \le 0. \end{cases} \tag{34}$$

For points in Ω_5, put $a = |X|/5$, $b = |Y|/5$. Because $a + b \le 1$, each input has only its central and sign-adjacent membership active. Their four joint min weights are

$$w_{00} = \min(1 - a, 1 - b) = 1 - \max(a, b),$$
$$w_{s0} = \min(a, 1 - b) = a,$$
$$w_{0s} = \min(1 - a, b) = b,$$
$$w_{ss} = \min(a, b). \tag{35}$$

The subscripts s denote the input's active sign label, not the scalar $s = |X| + |Y|$. Zero-input boundaries follow by continuity.

The denominator is therefore

$$D = 1 - \max(a, b) + a + b + \min(a, b) = 1 + 2\min(a, b) = 1 + 2m/5. \tag{36}$$

The central-central consequent is zero. The sign-central and central-sign rules contribute $(5/2)\operatorname{sgn}(X)a = X/2$ and $(5/2)\operatorname{sgn}(Y)b = Y/2$. For equal signs, the joint consequent is $5\operatorname{sgn}(X)$ and contributes $\operatorname{sgn}(X)m$; for opposing signs it is zero. The four quadrants therefore give

$$F(X, Y) = \begin{cases} \dfrac{F_0 + m}{1 + 2m/5}, & X > 0,\ Y > 0, \\ \dfrac{F_0 - m}{1 + 2m/5}, & X < 0,\ Y < 0, \\ \dfrac{F_0}{1 + 2m/5}, & X > 0,\ Y < 0, \\ \dfrac{F_0}{1 + 2m/5}, & X < 0,\ Y > 0. \end{cases} \tag{37}$$

On either axis $m = 0$, giving $F = F_0$. These cases establish the following identity, including boundaries by continuity.

Proposition 3.2 (Exact central surface). *For every* $(X, Y) \in \Omega_5$,

$$F(X, Y) = \frac{(X + Y)/2 + J(X, Y)}{1 + 2m/5}, \qquad J(X, Y) = \begin{cases} \operatorname{sgn}(X)m, & XY > 0, \\ 0, & XY \le 0. \end{cases} \tag{38}$$

This is an exact rule-level identity on Ω_5, not a fitted approximation. Outside this diamond, the cross weights in (**??**) need not equal a and b, so the formula must not be extended without a new case analysis.

3.7 Sign-conditioned response, sector and nonlinear geometry

3.7.1 Same-sign enhancement. When $XY > 0$, F_0 has magnitude $s/2$ and the same sign as X. Equation (**??**) gives

$$\frac{F}{F_0} = \frac{1 + 2m/s}{1 + 2m/5}. \tag{39}$$

Since $s \le 5$, its numerator is at least the denominator. Since $m \le s/2$, the numerator is at most 2. Thus $1 \le F/F_0 \le 2$. Define the normalized nonlinear residual by $\mathcal{R}_F(X, Y) = F(X, Y) - F_0(X, Y)$. Subtracting F_0 gives

$$\mathcal{R}_F = \frac{\operatorname{sgn}(X)m(1 - s/5)}{1 + 2m/5}, \qquad XY > 0. \tag{40}$$

For instance, $F(0.1, 0.1) = 0.2/1.04 \simeq 0.19230769$, compared with $F_0 = 0.1$. Both are surface values; the actual increment multiplies by negative K_u.

3.7.2 Opposing-sign attenuation and zero-output cases. When $XY < 0$, the interaction vanishes, so

$$F = \frac{F_0}{1 + 2m/5}, \qquad \mathcal{R}_F = -\frac{2m/5}{1 + 2m/5}F_0. \tag{41}$$

For $(X, Y) = (0.1, -0.05)$, $F_0 = 0.025$ and $F = 0.025/1.02 \simeq 0.024509804$. On either axis $m = 0$ and $F = F_0$. On the anti-diagonal $Y = -X$, both outputs are exactly zero; a ratio should not be evaluated there.

Corollary 3.3 (Regional pointwise sector). *For $(X, Y) \in \Omega_r$, $0 < r \le 5$, one can write*

$$F = \alpha(X, Y)F_0, \qquad \underline{\alpha}_r = \frac{1}{1 + r/5} \le \alpha(X, Y) \le 2. \tag{42}$$

Proof. Same-sign ratios lie in $[1, 2]$. For opposing signs with $F_0 \neq 0$, $m \le r/2$ gives $1/(1 + r/5) \le F/F_0 \le 1$. The axes agree with F_0. If $F_0 = 0$, both outputs vanish and any admissible α preserves the equality. □

The equivalent division-free inequality is

$$[F - \underline{\alpha}_r F_0][2F_0 - F] \ge 0. \tag{43}$$

In particular, Ω_1 gives $\alpha \in [5/6, 2]$, while Ω_5 gives $[1/2, 2]$. These bounds describe pointwise values of the specific nonlinear map. They are not a gain-scheduled stability proof; the closed loop must retain the input region and accommodate different node-dependent, time-varying α values.

3.7.3 Why an axis match is not a Jacobian linearization. The same-sign term J is positively homogeneous. For $t > 0$ and a fixed direction $v = (v_1, v_2)$, the central formula yields

$$F'(0; v) = \lim_{t \downarrow 0} \frac{F(tv_1, tv_2)}{t} = \frac{v_1 + v_2}{2} + J(v_1, v_2). \tag{44}$$

Consequently $F'(0; (1, 0)) = F'(0; (0, 1)) = 1/2$, but $F'(0; (1, 1)) = 2$. A linear derivative with the two axis values would instead give 1 on $(1, 1)$, a contradiction. The map is nonlinear and not Fréchet differentiable at the origin, despite exact agreement with F_0 on the coordinate axes. "Axis-calibrated linear counterpart" is therefore the appropriate term.

3.7.4 Where the central interaction comes from. As an analytic counterfactual, replace min rule weights by $w_{ab}^{\text{prod}} = \varphi_a(X)\varphi_b(Y)$, retaining the membership functions and singletons. On $|X|, |Y| \le 5$, the active consequents add their two single-input values and the denominator is $(\sum_a \varphi_a(X))(\sum_b \varphi_b(Y)) = 1$. Thus

$$F_{\text{prod}}(X, Y) = X/2 + Y/2 = F_0(X, Y). \tag{45}$$

The central nonlinear interaction is tied to normalized min activation, not merely to linguistic labels. The identity isolates the role of min activation; it does not assert linearity beyond the central square or an experimental comparison of product and min inference.

3.8 Nonlinear memory and what it does not identify

For identical scaled input pairs (X_k, Y_k), including consistent previous-error initialization, let A_k^{F} use F and A_k^0 use F_0, with equal initial accumulators. Subtracting their updates gives

$$A_k^{\text{F}} - A_k^0 = (A_{k-1}^{\text{F}} - A_{k-1}^0) + K_u \mathcal{R}_F(X_k, Y_k).$$

Telescoping yields

$$A_k^{\text{F}} - A_k^0 = K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{46}$$

With zero native/Fuzzy constructor states and the calibrated parameters, the same-input bookkeeping can also be written

$$A_k^{\text{F}} = k_p E_k + \kappa_I \sum_{\ell=0}^{k} E_\ell + K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{47}$$

The first two terms form a PI-like shadow evaluated on the Fuzzy trajectory's own inputs. The last term records accumulated nonlinear correction. If later inputs lie on the axes, the new residual is zero but the accumulated difference can remain nonzero.

This identity requires no accumulator projection/reset. Pure external clipping with retained raw states leaves the raw identity intact, but the applied difference must be evaluated through the limiter. Tracking/reset requires explicit additional state-change terms. Moreover, independently evolving PI and Fuzzy closed loops generate different measurements and possibly different physical event schedules. A same-input shadow identity therefore cannot be relabelled as their observed closed-loop difference.

3.9 Interpretation and the bridge to the stability section

The derivation provides implementation-consistent PI calibration, the exact central nonlinear map, and a regional sector suitable for a closed-loop certificate. Section **??** uses these properties after stating its plant, sampling, latency and initialization assumptions and proving that the sector's input region remains invariant. The present algebra does not change the continuous-clock Callisto plant into a discrete-time plant; it describes the controller updates embedded in that execution.

Same-sign scaled inputs indicate alignment of the two input channels. They do not always mean increasing absolute error: a transition from $e_{k-1} = -10$ to $e_k = 1$ has $e_k(e_k - e_{k-1}) > 0$ even though the absolute error decreased. Likewise, opposing-sign attenuation concerns a single increment relative to F_0, not an established network jitter attenuation property. Frequency disagreement, buffer excursion, common frequency and control activity must be evaluated separately.

4 Reference-Model Convergence and Buffer Margins

We connect the nonlinear controller of Section **??** to four separate properties: invariance of its analyzed input region, convergence of frequency and buffer disagreements, strict buffer-capacity safety, and positive physical oscillator frequency. We then establish a strict headroom comparison with its calibrated PI baseline for one specified initialization. The proof concerns an explicitly defined continuous fluid reference. Actual Callisto retains asynchronous local sampling, delivery delays and integer frame events, which are not silently removed from its experimental model.

4.1 Reference assumptions and continuous-time model

4.1.1 What the reference retains and idealizes. The reference retains the controller of Section **??**, continuous oscillator phases, local occupancy feedback, the native integrate-before-output PI convention and the proposed incremental normalized-min Fuzzy

surface. It assumes zero path delay, zero control-execution delay, fluid rather than integer occupancy and a common physical sampling grid $t_k = kH$, $H > 0$. A zero-order hold (ZOH) means that the applied frequency correction is constant on each interval $[t_k, t_{k+1})$. These are explicit analysis assumptions; local-clock polling in actual Callisto is not asserted to have this common grid. The equations sampled at kH below are exact endpoint relations for that continuous ZOH flow. They are neither numerical Euler integration nor a discrete firing-count replacement of the plant.

Let $\mathcal{N}_i^-$ be the fixed incoming-neighbour set and $d_i = |\mathcal{N}_i^-|$. The graph is connected, undirected and unit-weight, with a receiving channel in both directions. Its symmetric Laplacian L has diagonal d_i and off-diagonal -1 at neighbouring nodes. Each undirected connection contributes once to this Laplacian; the pair of directed receiving channels does not double its weights.

The physical phase is θ_i, the intrinsic frequency is ω_i^u, the applied correction is u_i, and the requested correction is q_i. The target occupancy of receiving edge $j \to i$ is $\beta^*_{j\to i}$. The fixed coefficient scale in the native PI law is $h = p/\omega_{\text{base}}$, where p is a local polling period and ω_{base} is the implementation reference frequency; H is the physical sampling period imposed only in this reference. The integral coefficient is $\kappa_I = k_i h$. These two time scales are kept distinct. All receiving buffers have common capacity $C > 0$, and the intrinsic frequencies ω_i^u are constant. The all-ones column vector is $\mathbf{1}$.

The transpose is denoted by $\top$. A symmetric matrix $A \succeq 0$ is positive semidefinite; $A \succ 0$ is positive definite. The matrix order $A \preceq B$ means $B - A \succeq 0$. The symbol $\otimes$ denotes the Kronecker product, and $\|z\|_\infty = \max_i |z_i|$.

4.1.2 Compatible occupancy targets and marking. Throughout the proved centered reference, all initial phases are equal. For zero delay the fluid law in the additive phase-offset convention is

$$\beta_{j\to i}(t) = \theta_j(t) - \theta_i(t) + \lambda_{j\to i}. \tag{48}$$

Here $\lambda_{j\to i}$ is the fixed additive phase offset. With equal initial phases it also equals the initial marking; this presentation does not identify these conventions for arbitrary unequal initial phases. The law follows the bittide fluid modelling framework [**? ?**]. Define $\beta^*_{j\to i} = \theta_j(0) - \theta_i(0) + \lambda_{j\to i}$, so the reference starts at its compatible targets. For a common constant ramp frequency ω_{ref} put

$$x_k = \theta(t_k) - \theta(0) - t_k \omega_{\text{ref}} \mathbf{1}. \tag{49}$$

The common ramp cancels from edge differences. Subtracting the target in (**??**) gives

$$\beta_{j\to i}(t_k) - \beta^*_{j\to i} = x_{j,k} - x_{i,k}, \qquad E_k = -Lx_k. \tag{50}$$

The vector E collects the Section **??** frame errors, actual minus target. Not every set of per-edge targets is compatible with a fixed marking and phase initialization; that compatibility is an assumption of this model.

For the worked example, equal initial phases and target 1024 imply $\lambda_{j\to i} = 1024$. The actual delayed runs have marking 1044 with 20 initial in-flight frames; simply retaining 1044 while imposing zero delay and the same centered occupancy would change the initialization.

4.2 From the original controller to the sampled recurrence

Define the intrinsic offsets $\delta\omega = \omega^u - \omega_{\text{ref}}\mathbf{1}$ and the physical frequency offsets $w_k = \delta\omega + u_k$. For $k \geq 1$, let $v_k = Hw_{k-1}$. The plant's previous genuine interval gives

$$x_k - x_{k-1} = Hw_{k-1} = v_k. \tag{51}$$

Indeed, on each zero-order-held interval,

$$x(t) = x_k + (t - t_k)w_k, \qquad t \in [t_k, t_{k+1}], \tag{52}$$

where $x(t) = \theta(t) - \theta(0) - t\omega_{\text{ref}}\mathbf{1}$. Integration gives $x_{k+1} = x_k + Hw_k$ exactly. The controller is updated only at the endpoints, but the plant evolves continuously throughout the interval.

The factor H is physical elapsed time; the controller factor h remains the native fixed coefficient scale. Section **??** uses $e_i = -\gamma_i E_i$, so the Fuzzy inputs are

$$X_k = K_e \Gamma L x_k, \qquad Y_k = K_{\Delta e} \Gamma L v_k, \qquad \Gamma = \operatorname{diag}(\gamma_1, \ldots, \gamma_n). \tag{53}$$

Here $K_e, K_{\Delta e}$ are chosen common for readability; node-dependent output scaling can still produce common calibrated k_p, k_i.

On $\Omega_r = \{(X, Y) : |X| + |Y| \leq r\}$, $0 < r \leq 5$, Section **??** gives $F = \alpha F_0$, where $F_0 = (X + Y)/2$ and $\underline{\alpha} = (1 + r/5)^{-1} \leq \alpha \leq \overline{\alpha} = 2$. With $D_{\alpha,k} = \operatorname{diag}(\alpha_{1,k}, \ldots, \alpha_{n,k})$, the calibrated increment becomes

$$u_k - u_{k-1} = -D_{\alpha,k}(\kappa_I L x_k + k_p L v_k). \tag{54}$$

Multiplication by H and the next plant interval yield

$$\begin{aligned} v_{k+1} &= v_k - D_{\alpha,k}(bLx_k + aLv_k), \\ x_{k+1} &= x_k + v_{k+1}, \qquad a = Hk_p, \quad b = H\kappa_I = Hk_i h. \end{aligned} \tag{55}$$

The extra H in b is essential: v is a scaled frequency. The surface factor $\alpha_{i,k}$ is node-dependent and state-dependent; freezing it and checking one constant-gain PI characteristic is insufficient.

4.2.1 The initial constructor call. The controller's stored previous error is not a previous plant interval. For the centered initialization $x_0 = 0$ with zero PI sample sum, zero Fuzzy accumulator and zero previous percentage error, the first original controller call gives $u_0 = 0$. Advancing $[0, H]$ then gives

$$x_1 = H\delta\omega, \qquad v_1 = H\delta\omega. \tag{56}$$

Only from $k = 1$ does (**??**) justify the recurrence. Nonzero initial occupancy error is outside this centered formulation: one must also introduce target-compatible phase coordinates or retain the affine error bias, then explicitly evaluate the original constructor and recheck the initial state and certificate. Evaluating the first call alone does not restore $E = -Lx$. The centered bootstrap is consistent with, but separate from, the event-level first-call PI/F_0 equivalence proved in Section **??**.

4.3 Disagreement projection and the symmetric sector matrix

Since $L\mathbf{1} = 0$, a common phase shift does not change occupancies. Choose U with orthonormal columns spanning the subspace orthogonal to $\mathbf{1}$, so $U^\top U = I_{n-1}$ and $U^\top \mathbf{1} = 0$. Choose its columns as graph eigenvectors, giving

$$L = U\Lambda U^\top, \qquad \Lambda = \operatorname{diag}(\mu_2, \ldots, \mu_n), \quad 0 < \mu_2 \leq \mu_n. \tag{57}$$

I_{n-1} denotes the identity matrix of dimension $n-1$. Define

$$\xi_k = \Lambda^{1/2} U^\top x_k, \qquad \eta_k = \Lambda^{1/2} U^\top v_k, \qquad y_k = (\xi_k^\top, \eta_k^\top)^\top. \quad (58)$$

The symbol ξ is projected phase and η already contains the frequency scale H. The coordinates retain the physical-time scale H; in particular, η is a scaled frequency disagreement rather than an integral-controller state.

For $M_k = \Lambda^{1/2} U^\top D_{\alpha,k} U \Lambda^{1/2}$, substitution in (??) gives

$$\begin{aligned} \eta_{k+1} &= -bM_k\xi_k + (I_{n-1} - aM_k)\eta_k, \\ \xi_{k+1} &= (I_{n-1} - bM_k)\xi_k + (I_{n-1} - aM_k)\eta_k. \end{aligned} \quad (59)$$

In particular, the diagonal D_α need not commute with L; the projection produces the symmetric M_k instead of assuming shared modes. Its quadratic form satisfies

$$\underline{\alpha}\Lambda \preceq M_k \preceq \overline{\alpha}\Lambda, \qquad \mathrm{eig}(M_k) \subseteq [\nu_-, \nu_+] = [\underline{\alpha}\mu_2, \overline{\alpha}\mu_n]. \quad (60)$$

The scalar ν used below is a possible eigenvalue of M, not a node frequency or the number of communication edges.

4.4 A common Lyapunov certificate from two endpoints

Define For matrices $M \in \mathbb{R}^{(n-1)\times(n-1)}$ and scalars ν, define

$$\mathcal{B}(M) = \begin{bmatrix} I - bM & I - aM \\ -bM & I - aM \end{bmatrix}. \quad (61)$$

The identity blocks have dimension $n-1$. Equations above give $y_{k+1} = \mathcal{B}(M_k) y_k$. A scalar block associated with an eigenvalue of M_k is

$$B(\nu) = \begin{bmatrix} 1 - b\nu & 1 - a\nu \\ -b\nu & 1 - a\nu \end{bmatrix}. \quad (62)$$

For a symmetric positive-definite matrix $\bar{P} \in \mathbb{R}^{2\times 2}$, put $P = \bar{P} \otimes I_{n-1}$. The scalar $V_k = y_k^\top P y_k$ is a quadratic measure of phase and frequency disagreement, not of absolute phase.

Lemma 4.1 (Endpoint certificate). *Suppose* $0 < \rho < 1$ *and*

$$\begin{aligned} B(\nu_-)^\top \bar{P} B(\nu_-) &\preceq \rho \bar{P}, \\ B(\nu_+)^\top \bar{P} B(\nu_+) &\preceq \rho \bar{P}. \end{aligned} \quad (63)$$

Then every update whose inputs satisfy the regional sector obeys

$$V_{k+1} \le \rho V_k. \quad (64)$$

The conclusion permits node-dependent and time-varying sector factors.

Proof. First consider one possible scalar eigenvalue. For $\nu = (1-t)\nu_- + t\nu_+$, $0 \le t \le 1$, affine dependence gives $B(\nu) = (1-t)B_- + tB_+$, where $B_\pm = B(\nu_\pm)$. Expanding the two quadratic expressions yields

$$\begin{aligned} &(1-t)B_-^\top \bar{P} B_- + tB_+^\top \bar{P} B_+ - B(\nu)^\top \bar{P} B(\nu) \\ &\quad = t(1-t)(B_+ - B_-)^\top \bar{P} (B_+ - B_-) \succeq 0. \end{aligned} \quad (65)$$

Together with the endpoint assumptions, this proves $B(\nu)^\top \bar{P} B(\nu) \preceq \rho \bar{P}$ throughout the continuous interval $[\nu_-, \nu_+]$. No eigenvalue grid is required.

Next, each M_k is real symmetric, so it admits an orthogonal factorization $M_k = Q_k \,\mathrm{diag}(\nu_{1,k}, \dots, \nu_{n-1,k}) Q_k^\top$. Rotating both ξ_k and η_k by $Q_k^\top$ reduces the dynamics to the scalar blocks $B(\nu_{\ell,k})$. The corresponding block rotation leaves $P = \bar{P} \otimes I_{n-1}$ unchanged. Thus summing the scalar inequalities gives $\mathcal{B}(M_k)^\top P \mathcal{B}(M_k) \preceq \rho P$, which is (??). Because the witness P is the same for every permitted M_k, Q_k may change at every step. No simultaneous diagonalization of the node-dependent matrices is assumed. □

The remaining task is to show that the actual nonlinear inputs stay in the region where the sector was obtained. Stability of a family of hypothetical linear blocks alone would not establish that fact.

4.5 Closing the invariant input region

Let $\mathbf{e}_i$ denote the ith unit vector and define the row $a_i^\top = \mathbf{e}_i^\top U \Lambda^{1/2}$. Its squared Euclidean norm is $\|a_i\|^2 = L_{ii} = d_i$. The Fuzzy inputs become

$$X_{i,k} = \gamma_i K_e a_i^\top \xi_k, \qquad Y_{i,k} = \gamma_i K_{\Delta e} a_i^\top \eta_k. \quad (66)$$

For any vector c and $P \succ 0$, Cauchy–Schwarz in the quadratic metric gives $(c^\top y)^2 \le (y^\top P y)(c^\top P^{-1} c)$. Also $|X| + |Y| = \max_{\sigma=\pm 1} |X + \sigma Y|$. Applying that output bound on $V \le R$ yields

$$\begin{aligned} |X_{i,k}| + |Y_{i,k}| &\le \sqrt{R\Psi_i}, \\ \Psi_i &= \gamma_i^2 d_i \max_{\sigma=\pm 1} [K_e, \sigma K_{\Delta e}] \bar{P}^{-1} [K_e, \sigma K_{\Delta e}]^\top. \end{aligned} \quad (67)$$

The scalar Ψ_i is the largest squared input gain on the energy ellipsoid. If $V_1 \le R$ and $R\Psi_i \le r^2$ for every node, then: each node's first scaled input pair belongs to Ω_r; the valid sector gives $V_2 \le \rho V_1 \le R$; the same output bound keeps the next inputs in Ω_r; induction repeats. This proves the sector domain and disagreement ellipsoid invariant together, rather than assuming either.

Equivalently, a valid radius must satisfy

$$V_1 \le R \le \min_i \frac{r^2}{\Psi_i}. \quad (68)$$

The initialization, normalization through d_i and C, the Fuzzy input scales and the graph-dependent witness therefore enter the same condition. When $\Psi_i = 0$, that node places no restriction on R; otherwise the displayed quotient is finite. The inference surface is still nonlinear inside the invariant region. The sector is an enclosure of its instantaneous response, not an assertion that its gain is fixed.

Because $P \succ 0$ and $V_k \le \rho^{k-1} V_1$, the projected phase and frequency decay exponentially. The ellipsoid constrains disagreement; the common physical phase continues to grow and is not bounded by V. Thus convergence means agreement of rates and restoration of compatible occupancy targets, not convergence of the absolute oscillator phases to constants.

4.6 From disagreement to physical buffer and clock bounds

4.6.1 Edge occupancy and intersample time. For edge $e = (j \to i)$, define its signed unit incidence vector $\mathbf{b}_e = \mathbf{e}_j - \mathbf{e}_i$. Since it is orthogonal to $\mathbf{1}$, its output from the projected phase is

$$\mathbf{b}_e^\top x_k = \mathbf{b}_e^\top U \Lambda^{-1/2} \xi_k.$$

Define the graph pseudoinverse $L^\dagger = U \Lambda^{-1} U^\top$ and effective resistance $R_e^{\mathrm{eff}} = \mathbf{b}_e^\top L^\dagger \mathbf{b}_e$. The quadratic output inequality gives

$$|\beta_e(t_k) - \beta_e^*|^2 \le V_k [\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}}. \quad (69)$$

Under ZOH the phase, hence each fluid occupancy, is affine on an interval. For $0 \le s \le 1$ its deviation is a convex combination of endpoint deviations, so its absolute value cannot exceed the larger

endpoint absolute value. This covers all physical time, including the initial interval from the zero state to x_1. It would not apply unchanged to arbitrary delayed integer arrival events.

Corollary 4.2 (Buffer safety within the reference model). *Suppose the compatible centered initialization and the invariant-domain conditions above hold with $V_1 \le R$. For each receiving edge define*

$$m_e = \min\{\beta_e^*, C - \beta_e^*\},$$
$$B_e(R) = \sqrt{R[\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}}}. \tag{70}$$

If $0 < \beta_e^* < C$ and $B_e(R) < m_e$ on every receiving edge, then the unconstrained fluid execution satisfies $0 < \beta_e(t) < C$ for all $t \ge 0$. In addition, $\beta_e(t) - \beta_e^* \to 0$ as $t \to \infty$.

Proof. At t_k, the squared deviation is bounded by $V_k[\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}}$ and hence by $B_e(R)^2$. The initial deviation is zero. On $[t_k, t_{k+1}]$, the deviation is a convex combination of its endpoint deviations, so its absolute value is at most $B_e(R)$. The strict headroom inequalities give the lower and upper capacity inequalities. Finally $V_k \le \rho^{k-1} V_1 \to 0$ and the same interpolation argument imply convergence at every physical time, not only at samples. □

Here $B_e(R) < m_e$ is a sufficient condition. If it fails, the certificate is inconclusive; failure of this conservative test does not prove a buffer overflow or underflow. The condition also concerns the unconstrained flow. It does not identify a protection-induced stall with successful control convergence.

4.6.2 Frequency-difference safety and convergence. Define the physical frequency spread by

$$\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t). \tag{71}$$

A small frequency spread means close oscillator rates; it does not require the phases or their integer counts to coincide. For any pair $i \ne j$, not necessarily adjacent, define

$$R_{ij}^{\mathrm{eff}} = (\mathbf{e}_i - \mathbf{e}_j)^\top L^\dagger (\mathbf{e}_i - \mathbf{e}_j), \qquad R_{\max}^{\mathrm{eff}} = \max_{i \ne j} R_{ij}^{\mathrm{eff}}, \tag{72}$$

where $\mathbf{e}_i$ is the ith unit vector. Connectedness makes these quantities finite. During $[t_k, t_{k+1})$, $\omega_i(t) = \omega_{\mathrm{ref}} + w_{i,k}$, while $v_{k+1} = H w_k$. Therefore

$$\begin{aligned} |\omega_i(t) - \omega_j(t)|^2 &= H^{-2} |(\mathbf{e}_i - \mathbf{e}_j)^\top v_{k+1}|^2 \\ &\le H^{-2} V_{k+1} [\bar{P}^{-1}]_{22} R_{ij}^{\mathrm{eff}} \\ &\le H^{-2} R \rho^k [\bar{P}^{-1}]_{22} R_{ij}^{\mathrm{eff}}. \end{aligned} \tag{73}$$

The factor V_{k+1}, rather than V_k, is essential: v_k represents the frequency held during the previous interval. The case $k = 0$ is covered by the explicitly constructed bootstrap state y_1. At switching times the same bound applies to the newly held frequency; the previous interval bounds the left-hand limit.

Corollary 4.3 (A prescribed frequency-spread threshold). *Under the invariant-domain conditions, let $\Delta_{\mathrm{safe}} > 0$ be a specified engineering tolerance on pairwise frequency differences. If*

$$D(R) := \frac{1}{H} \sqrt{R[\bar{P}^{-1}]_{22} R_{\max}^{\mathrm{eff}}} < \Delta_{\mathrm{safe}}, \tag{74}$$

then for every $t \in [t_k, t_{k+1})$, $k \ge 0$,

$$\Delta_\omega(t) \le D(R) \rho^{k/2} < \Delta_{\mathrm{safe}}, \qquad \Delta_\omega(t) \longrightarrow 0. \tag{75}$$

Proof. The spread equals $\max_{i,j} |\omega_i - \omega_j|$. Maximizing (**??**) and taking square roots proves the bound. Since $0 < \rho < 1$, the bound tends to zero. □

The threshold is a user-specified performance or hardware tolerance; logical synchrony does not by itself prescribe its numerical value. This proof bounds the transient as well as the limiting disagreement. It neither proves a smaller Fuzzy frequency peak than PI nor adds a guarantee for arbitrary delayed or stochastic executions. Frequency spread also cannot certify positive absolute frequency: a collection of equal nonpositive rates would have zero spread. The common frequency must therefore be bounded separately.

4.6.3 The unprojected common frequency. Let $\bar{w}_k = n^{-1} \mathbf{1}^\top w_k$ be the common-frequency offset. Put $\alpha_c = (\overline{\alpha} + \underline{\alpha})/2$ and $s_\alpha = (\overline{\alpha} - \underline{\alpha})/2$. With $g = (\kappa_I, k_p)^\top$, averaging (**??**) and using $\mathbf{1}^\top U = 0$ gives

$$\bar{w}_k - \bar{w}_{k-1} = -\frac{1}{n} \mathbf{1}^\top (D_{\alpha,k} - \alpha_c I_n) U \Lambda^{1/2} (\kappa_I \xi_k + k_p \eta_k). \tag{76}$$

The second component of g is k_p, not $H k_p$, since η already contains H. Bounding each factor yields

$$|\bar{w}_k - \bar{w}_{k-1}| \le s_\alpha \sqrt{\frac{\mu_n}{n} g^\top \bar{P}^{-1} g} \sqrt{V_k}. \tag{77}$$

The geometric series in $\sqrt{\rho}$ is summable. Hence

$$\begin{aligned} \sup_k |\bar{w}_k| &\le |\bar{w}_0| + \frac{s_\alpha \sqrt{(\mu_n/n) R g^\top \bar{P}^{-1} g}}{1 - \sqrt{\rho}} \\ &\le |\bar{w}_0| + \frac{2 s_\alpha}{1 - \rho} \sqrt{\frac{\mu_n}{n} R g^\top \bar{P}^{-1} g} = W_c. \end{aligned} \tag{78}$$

The last inequality follows from $(1 - \sqrt{\rho})^{-1} = (1 + \sqrt{\rho})/(1 - \rho) \le 2/(1 - \rho)$. Absolute summability also proves convergence of $\bar{w}_k$. Combined with vanishing disagreement, all node frequencies converge to one value $\omega_\infty = \omega_{\mathrm{ref}} + \lim_k \bar{w}_k$. For Fuzzy PI that limit need not be the initial intrinsic mean.

4.6.4 Closing actuator saturation and positive frequency. The ith disagreement-frequency output satisfies

$$|w_{i,k} - \bar{w}_k| \le W_d, \qquad W_d = \max_i \frac{1}{H} \sqrt{R[\bar{P}^{-1}]_{22} (L^\dagger)_{ii}}. \tag{79}$$

Applying the state bound to y_{k+1} covers w_k; y_1 covers w_0. As $u = w - \delta\omega$, we obtain

$$|u_{i,k}| \le W_c + W_d + \|\delta\omega\|_\infty, \qquad \omega_i(t) \ge \omega_{\mathrm{ref}} - W_c - W_d. \tag{80}$$

First derive these bounds for the unconstrained trajectory. If the first is strictly below $u_{\max}$ and the second is positive, the limiter is never reached, so the constrained execution coincides with that trajectory. This continuation argument closes the assumption of no clipping. The buffer bound must likewise fit within physical capacity; protecting a buffer by blocking execution is a different outcome.

Together, the following sufficient conditions connect the controller and network parameters to the required properties:

$$\begin{aligned}
&V_1 \le R, \qquad R\Psi_i \le r^2 \quad \text{for every } i,\\
&0 < \beta_e^* < C \quad \text{for every } e,\\
&R[\bar{P}^{-1}]_{11} R_e^{\text{eff}} < m_e^2 \quad \text{for every } e,\\
&R[\bar{P}^{-1}]_{22} R_{\max}^{\text{eff}} < H^2 \Delta_{\text{safe}}^2,\\
&W_c + W_d + \|\delta\omega\|_\infty < u_{\max},\\
&\omega_{\text{ref}} - W_c - W_d > 0,
\end{aligned} \tag{81}$$

provided the endpoint inequalities of Lemma **??** hold. The frequency-spread inequality is required only when a specific threshold is an objective; convergence alone does not require choosing that threshold. All quantities are evaluated for one consistent model, initialization and parameter set. In particular, changing H, h, C, node degree, controller gains or topology can change more than one inequality. These are sufficient certification tests, not a characterization of the largest safe set or necessary stability boundaries.

4.7 The prescribed triangle certificate

Take the three-node complete graph, giving $\mu_2 = \mu_3 = 3$, $R_e^{\text{eff}} = 2/3$ and $(L^\dagger)_{ii} = 2/9$. The parameters are

$$\begin{aligned}
H = h = \frac{30000000}{29999}, \quad & k_p = \frac{1}{500000}, \quad & k_i = \frac{1}{20000000000},\\
K_e = \frac{1}{10}, \quad & K_{\Delta e} = \frac{29999}{7500}, \quad & K_u = -\frac{768}{18749375},\\
C = 2048, \quad & \beta_e^* = \beta_e(0) = 1024, \quad & \lambda_e = 1024.
\end{aligned} \tag{82}$$

Frequencies in this numerical example are normalized, and physical time is measured in reference ticks. Use $\omega^u = (1.0002, 0.9998, 0.9999)^\top$ and its average as $\omega_{\text{ref}} = 29999/30000$, so $\delta\omega = (7, -5, -2)^\top/30000$ and $\bar{w}_0 = 0$. At the compatible startup (**??**), use

$$\bar{P} = \frac{1}{50000}\begin{bmatrix} 9 & 181 \\ 181 & 49629 \end{bmatrix}, \quad \rho = \frac{1599}{1600}, \quad r = 1. \tag{83}$$

Here $\det \bar{P} = 4139/25000000 > 0$ and the leading diagonal entry is positive. Since the sum of its four entries is one,

$$R_0 = V_1 = H^2 \delta\omega^\top L \delta\omega = \frac{234000000}{899940001}. \tag{84}$$

This exact fraction is used in all inequalities. The displayed approximation 0.2600173342 is slightly below the exact value and is not substituted as an upper energy radius.

The sector is $[5/6, 2]$, hence the spectral endpoints are $5/2$ and 6. For $D_\nu = \rho\bar{P} - B(\nu)^\top \bar{P} B(\nu)$, exact rational arithmetic verifies positive leading minor and determinant at both ends:

ν	$[D_\nu]_{11}$ (approx.)	$\det D_\nu$ (approx.)
5/2	$8.2193625299 \times 10^{-7}$	$7.8876135318 \times 10^{-11}$
6	$2.0776400066 \times 10^{-6}$	$7.3380136479 \times 10^{-9}$

The accompanying machine-readable certificate stores the exact fractions; the decimals are only a readable display, not rounded tests of positivity. Using R_0 gives the following conservative, outward bounds:

Quantity	Certified bound
$\sup_{i,k}(\lvert X_{i,k}\rvert + \lvert Y_{i,k}\rvert)$	$< 0.171455 < 1$
Coarse edge excursion	< 32.237442 frames
$\sup_k \lvert\bar{w}_k\rvert$	< 0.003682181
$\sup_{i,k} \lvert w_{i,k} - \bar{w}_k\rvert$	< 0.000250634
$\sup_{i,k} \lvert u_{i,k}\rvert$	$< 0.004166148 < 0.01$
$\inf_{i,t} \omega_i(t)$	> 0.99603385

The coarse excursion already fits within the 1024-frame lower and upper headroom. The frequency bounds use the exact rational-square-root enclosures, not a sum of prematurely rounded display values.

4.8 The exact PI all-time baseline

For $J \in \{\text{PI}, \text{F}\}$ define

$$B_J = \sup_{t \ge 0} \max_e |\beta_e^J(t) - 1024|. \tag{85}$$

The supremum varies time and receiving edges, while the initialization and intrinsic-frequency vector remain fixed. For the same plant, initialization and gains, native PI has $D_\alpha = I_n$. Averaging its increments shows that the mean frequency is preserved. On the triangle, both disagreement modes have eigenvalue 3. Define $\chi_0 = 0$, $\chi_1 = 1$ and

$$\chi_{k+1} = (2 - 3a - 3b)\chi_k - (1 - 3a)\chi_{k-1}. \tag{86}$$

To derive it, use $v_k = x_k - x_{k-1}$ in the PI state recurrence:

$$x_{k+1} = (2I_n - (a+b)L)x_k - (I_n - aL)x_{k-1}.$$

Since $L\delta\omega = 3\delta\omega$ and $x_1 = H\delta\omega$, induction gives $x_k = H\chi_k\delta\omega$. With $\Delta_0 = \max_i \omega_i^u - \min_i \omega_i^u = 1/2500$,

$$\max_e |\beta_e(t_k) - 1024| = H\Delta_0|\chi_k|. \tag{87}$$

This exact rational sequence is enumerated through $N = 200$. Its unique prefix maximum occurs at $k = 111$, with

$$23.40370130812701 \le \max_{0 \le k \le N} H\Delta_0|\chi_k| \le 23.40370130812702. \tag{88}$$

It is not yet an all-time assertion until the tail is bounded.

At $N = 200$ put $\delta\chi_N = \chi_N - \chi_{N-1}$, the scalar response difference. The exact state energy is

$$V_N = R_0\{\bar{P}_{11}\chi_N^2 + 2\bar{P}_{12}\chi_N\delta\chi_N + \bar{P}_{22}(\delta\chi_N)^2\}. \tag{89}$$

The common certificate also covers PI's scalar value $\nu = 3$. Thus for all $k \ge N$, the remaining buffer excursion is at most $\sqrt{(2/3)[\bar{P}^{-1}]_{11} V_N}$, with an outward upper bound 17.95805698665804 frames. This is strictly below the attained prefix maximum. ZOH interpolation therefore makes (**??**) the exact all-physical-time B_{PI} enclosure, reached at time $111H$.

4.9 The Fuzzy interval prefix and contracting tail

A direct floating-point trajectory can locate a candidate peak but does not enclose the exact one. The certificate instead propagates intervals for all components of x_k and v_k on a grid $S^{-1} = 10^{-60}$. A rational number q_{rat} is enclosed by

$$\left[\frac{\lfloor Sq_{\text{rat}}\rfloor}{S}, \frac{\lceil Sq_{\text{rat}}\rceil}{S}\right].$$

Table 2: Certified excursions in the prescribed fluid reference (frames).

Controller	Prefix peak enclosure	Tail upper bound
PI	[23.40370130812701, 23.40370130812702]	17.95805698665804
Fuzzy	[17.314056817962, 17.314056817963]	12.29271005372874

Each multiplication uses the extrema of the four endpoint products, rounded outward; division uses a strictly positive enclosed denominator and outward-rounded endpoint quotients. Addition and subtraction use endpoint arithmetic. These operations use integers and rational comparisons, not floating-point rounding assumptions.

For the central formula, $m = \min(|X|, |Y|)$ is enclosed by interval absolute values and endpoint minima. Its interaction term can be written $J = \min(X, Y)$ for two positive inputs, $J = \max(X, Y)$ for two negative inputs and zero otherwise. It is coordinatewise nondecreasing, including across the axes; its lower and upper corner values enclose it on any input rectangle. The positive denominator $1 + 2m/5$ then permits a valid interval evaluation of the exact map. The invariant-domain proof ensures that this central formula is applicable to the exact trajectory.

Propagating the genuine bootstrap state through state $N = 200$ gives a prefix maximum enclosed by

$$[17.314056817962,\ 17.314056817963] \quad \text{frames.} \tag{90}$$

The prefix retains all node coordinates, so common drift is not silently discarded. On the triangle the quadratic energy is equivalently

$$V_k = \sum_{i<j} \left\{ \bar{P}_{11}(x_{i,k} - x_{j,k})^2 + 2\bar{P}_{12}(x_{i,k} - x_{j,k})(v_{i,k} - v_{j,k}) + \bar{P}_{22}(v_{i,k} - v_{j,k})^2 \right\} \tag{91}$$

Evaluating this expression outward gives $\overline{V}_N \geq V_N$. For all later samples,

$$|\beta_e(t_k) - 1024|^2 \leq \frac{2}{3}[\bar{P}^{-1}]_{11}\rho^{k-N}\overline{V}_N. \tag{92}$$

For either controller, an equivalent tail bound that makes the finite-prefix/infinite-tail decomposition explicit is

$$T_N = \sqrt{\frac{2}{3}[\bar{P}^{-1}]_{11}\overline{V}_N}. \tag{93}$$

For PI one can take $\overline{V}_N = V_N$ exactly; for Fuzzy, $\overline{V}_N$ is an outward interval upper bound. Once T_N is below the certified prefix lower bound, no later sample can exceed the prefix maximum. This closes the infinite time horizon without selecting an arbitrarily long simulation duration.

An outward upper bound for this whole tail is 12.29271005372874 frames, strictly below the prefix lower bound. Including $x_0 = 0$ and intersample interpolation gives $B_F < 17.315$ on the entire physical time axis. The exact Fuzzy peak is more tightly enclosed by (**??**); the rounded 17.315 value is used for a simple, conservative comparison.

The enclosures used in the comparison are collected in Table **??**. The prefix intervals are outward enclosures of actual peak values; the tail entries are upper bounds. Their different roles are essential for the strict comparison.

4.10 Strict comparison, engineering meaning and scope

Theorem 4.4 (Strict headroom benefit for a prescribed initialization). *For the triangle, parameters and compatible initialization above, initialize the PI sample sum, the Fuzzy accumulator and the stored previous percentage error to zero. Both reference executions keep positive physical frequencies, remain strictly inside the buffer interval* $(0, 2048)$ *and never activate the* 0.01 *correction limiter. Buffer deviations and all pairwise frequency differences converge to zero. Their all-time excursions satisfy*

$$B_F < 17.315 < 23.40370130812701 \leq B_{PI}. \tag{94}$$

For the minimum lower/upper headroom $M_J = 1024 - B_J$,

$$M_F > 1006.685, \qquad M_F - M_{PI} > 6.08870130812701 \quad \textit{frames.} \tag{95}$$

Proof. The endpoint inequalities and the radius test certify the invariant nonlinear input region. The buffer, correction and absolute-frequency bounds establish capacity safety, absence of clipping and positive frequencies. The invariant disagreement energy contracts, so the buffer and frequency-difference conclusions follow. PI is included by taking $\alpha_i = 1$, which belongs to the same sector; PI also preserves its initial mean frequency. The certified finite-prefix enclosures and the tail bounds imply that both all-time excursions are attained in the prefix. The Fuzzy prefix upper bound is below 17.315, whereas PI attains the enclosed peak at sample 111. Intersample affine occupancy rules out a missed larger fluid peak. Finally, each edge has a reverse edge with the negative occupancy deviation, so the worst lower/upper headroom is exactly $1024 - B_J$. Subtracting the strict excursion bounds gives the stated gain in headroom. □

The comparison uses a certified Fuzzy upper bound and an attained PI peak. Merely comparing two conservative upper bounds would not establish the strict ordering of the actual excursions.

Both reserves are already large. The comparison establishes a conditional resource-margin benefit, not a demonstrated storage bottleneck or hardware cost reduction at this operating point.

The supremum is over time and edges for one prescribed intrinsic-frequency vector. It is not an expanded safe set for all 400-ppm patterns, a guarantee against arbitrary latency perturbations or superiority over independently tuned PI. For a different capacity, $\gamma_i = 100/(d_i C)$ changes; keeping the Fuzzy gains unchanged generally changes the token-domain nonlinear map. A new capacity certificate requires a fresh derivation or a verified map-preserving normalization change. Physical occupancy constraints also do not replace the delayed runtime's in-flight and credit/readiness rules.

This is a computer-assisted mathematical certificate with reproducible integer/rational checks. It is not proof-assistant verification of the Julia implementation. The actual asynchronous, delayed, quantized cases belong in the experimental section. This division preserves the distinction between a theorem about the declared reference and empirical evidence about Callisto.

5 Evaluation

The evaluation addresses three questions. First, does the implementation respect the PI calibration and initialization derived in

Section ??? Second, are the reference-model calculations consistent with the implemented nonlinear rule law? Third, how does the full Fuzzy controller compare with fixed PI baselines under post-centering delivery disturbances in continuous-phase Callisto? These questions require separate evidence: implementation equivalence, reference-model consistency, and finite-horizon performance do not establish the same claim.

5.1 Evidence groups and comparison design

We use three completed actual-Callisto campaigns, denoted M0, M2 and M3, and separate fluid-reference calculations. M0 comprises six PI/linear- counterpart pairs, or 12 actual runs. M2 comprises four actual runs: PI and full Fuzzy, each with a clean trace and a single-edge latency burst. M3 comprises 180 actual validation runs: 30 independent paired seeds, two random-latency regimes and three fixed controllers. Six M3 pilot runs are separate from the 180-run validation and are not treated as independent test replications. Reference calculations are not added to the actual-Callisto run count.

The primary performance comparison is moderate PI against the full Fuzzy law calibrated to its PI gain coordinates. M3 additionally includes a prespecified stronger PI candidate. All candidates are frozen within these campaigns. The linear surface F_0 is used only for the M0 implementation check and explicitly labeled same-input diagnostics; M2 and M3 execute the full 25-rule nonlinear law. Axis matching controls scales and update conventions, but is not an assertion that either candidate is the best independently tuned controller.

5.2 Continuous Callisto setup and delivery disturbances

M2 and M3 use the continuous oscillator phases, local-clock scheduling and integer receiving-buffer counts described in Section ??. Table ?? gives their common configuration. Reported token counts represent counted frames in this model. There is no shared physical controller grid in these actual runs. The fixed native integration scale is $h = p/\omega_{\text{base}}$, with the same convention in the calibrated Fuzzy controller. The initial intrinsic-frequency spread is 400 ppm of the normalized frequency unit.

Burst after nominal centering. In M2, desired send-time latency on one directed channel, identified as $1 \to 2$ in the runtime configuration, changes from 20 to 40 during $[2 \times 10^6, 2.1 \times 10^6)$ and returns to 20 afterwards. All other channels remain at 20. Each controller has its own clean counterfactual with latency 20 throughout. A controller's burst and clean histories match before onset, so their subsequent difference isolates the effect of the delivery disturbance for that controller. The nominal states of PI and Fuzzy are not required to be identical to each other.

Bounded random latency. M3 uses fixed physical profile-update grids $T_c = 1000$ and 10000, independently of the unchanged controller polling period. For each generated link profile the bounded recursion is

$$z_0 = 5\varepsilon_0,$$
$$z_m = r_z z_{m-1} + (1 - r_z)5\varepsilon_m, \qquad r_z = e^{-1}, \quad \varepsilon_m \sim \mathcal{U}[-1, 1], \tag{96}$$

with desired delay $20 + z_m \in [15, 25]$ after onset. Here m indexes profile updates, not controller measurements. The designation T_c refers to the profile grid, not an exactly measured correlation time. Before onset, including negative-time prehistory, delay is 20. Seeds 20001–20030 are independent validation replications; pilot seed 1001 is excluded. For each seed and regime, all three controllers receive identical externally generated profile content.

Both disturbance families use the declared causal running-maximum arrival envelope. Frame order is retained and a delay decrease can release an ordered batch. No finite serialization-rate model, frame dropout or application-payload experiment is included. These are specified transport stress models rather than claims about arbitrary network traffic.

5.3 Metrics, windows and finite-run admissibility

Disagreement and absolute frequency. For the instantaneous common frequency $\bar{\omega}(t) = n^{-1} \sum_i \omega_i(t)$, define

$$\sigma_\omega(t) = \sqrt{\frac{1}{n} \sum_i [\omega_i(t) - \bar{\omega}(t)]^2},$$
$$s(t) = 10^6 \sigma_\omega(t)/\omega_{\text{unit}}, \qquad \omega_{\text{unit}} = 1. \tag{97}$$

Thus s is an instantaneous across-node root-mean-square (RMS) amplitude in parts per million (ppm). Its normalization is the fixed unit frequency, not ω_{base} or the reference ramp frequency used in Section ??. For $W = [t_s, t_e)$ with duration $T_W = t_e - t_s$, report

$$R_W = \sqrt{T_W^{-1} \int_W s(t)^2 \, dt},$$
$$I_W = \int_W s(t) \, dt, \qquad P_W = \sup_{t \in W} s(t). \tag{98}$$

The recorded integral absolute error (IAE) is I_W, the integral of this nonnegative RMS amplitude. It is not the integral of the mean absolute node deviation. Moments are integrated over complete held-frequency intervals using physical-duration weights, not uniform averages of controller updates.

These quantities must also be distinguished from the pairwise spread $\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t)$ of Section ??. Cauchy–Schwarz gives $\Delta_\omega(t) \le \sqrt{2n}\, \sigma_\omega(t)$, but a window RMS does not bound a pointwise transient by itself. We retain positive- frequency checks and common-frequency diagnostics separately. For M2, a supplementary descriptive spread calculation uses the already saved shared-time frequency intervals; it is not a new simulation or a preregistered primary endpoint.

Buffer excursion and control activity. Measured buffer excursion and remaining headroom are

$$B_W^{\text{obs}} = \max_e \sup_{t \in [t_s, t_e]} |\beta_e(t) - \beta_e^*|,$$
$$H_W^{\text{obs}} = \min_e \inf_{t \in [t_s, t_e]} \min\{\beta_e(t), C - \beta_e(t)\}. \tag{99}$$

Every integer phase crossing and ordered arrival is reconstructed. Simultaneous events are grouped under the frozen convention, and buffer extrema include the right-endpoint state. Display samples are not the source of peak or moment statistics. Conservative reference

Table 3: Common configuration of the aligned burst and random-latency experiments. Time is measured in physical reference ticks unless specified as local ticks.

Parameter	Value
Network	Three-node bidirectional triangle; six receiving buffers
Intrinsic frequencies ω^u	$(1.0002, 0.9998, 0.9999)^\top$
Initial phases; implementation base frequency	0.1 at every node; $\omega_{\text{base}} = 29999/30000$
Capacity; initial occupancy; target	$C = 2048$; $\beta_e(0) = \beta_e^* = 1024$ frames
Initial in-flight frames; delayed-run marking	20 per channel; 1044 in the recorded convention
Nominal physical path delay	20
Polling; control-execution delay	$p = 1000$; $d_{\text{ctrl}} = 10$ local ticks
Moderate PI (k_p, k_i)	$(2 \times 10^{-6}, 5 \times 10^{-11})$
Stronger fixed PI, M3 only	$(4 \times 10^{-6}, 5 \times 10^{-11})$
Full Fuzzy $(K_e, K_{\Delta e}, K_u)$	$(0.1, 3.9998666667, -4.0961365379 \times 10^{-5})$
Execution policy	$u_{\max} = 0.01$; shared guards enabled; reserve zero; no anti-windup
Optional mechanisms	Observer and supervisory mode switching disabled
Horizon; disturbance onset	4×10^6; $t_{\text{on}} = 2 \times 10^6$

certificates are not substituted for measured extrema. Mean per-node command total variation (TV) is

$$\mathrm{TV}_W = \frac{1}{n} \sum_i \sum_{k:\, t_s \le t_{i,k}^a \le t_e} |u_{i,k} - u_{i,k-1}|. \tag{100}$$

Activity uses inclusive endpoints in the frozen protocol; adjacent subwindow activities must not be added without handling their shared boundary. All evaluated M2/M3 runs have zero recorded saturation, so requested and applied command increments coincide. TV is a command- activity measure, not measured hardware energy or computational cost.

Windows and recovery. The primary disturbance window is $[2 \times 10^6, 4 \times 10^6)$. M2 additionally separates burst ON and post-OFF intervals at $t_{\text{off}} = 2.1 \times 10^6$. Before onset, both M2 controllers and all M3 cases pass three consecutive 10^5-tick nominal windows: disagreement RMS at most 2 ppm, each buffer's absolute mean bias at most two frames and standard deviation at most one frame, common-frequency temporal standard deviation at most 1 ppm and three-window mean range at most 1 ppm, with positive frequencies/progress and no intervention. These are finite engineering centering criteria, not exact equilibrium.

M2 component recovery requires these criteria in three consecutive post-OFF windows. Strict recovery additionally requires the window-mean common frequency to return within 2 ppm of that controller's own clean trajectory over the same physical-time window. A missing qualifying return by 4×10^6 is right-censored; it is neither zero recovery time nor proof of permanent nonrecovery. M3 uses $[3.6 \times 10^6, 4 \times 10^6)$ for tail diagnostics. The separate M2 memory decomposition below uses $[3.7 \times 10^6, 4 \times 10^6)$.

5.4 Implementation equivalence and reference consistency

M0 varies polling over 100 and 1000 local ticks, initial frequency spread over 35 and 400 ppm, and includes capacity/target settings $(2048, 1024)$, $(512, 128)$ and $(512, 384)$ across its six matched pairs. All pairs use constant path delay 20, execution delay 10 and a 2×10^5-tick horizon. Target-centered percentages are used for non-midpoint targets. Across 19,800 paired updates, the maximum requested/applied correction difference is 5.29×10^{-18}. Physical and data phases coincide at all evaluated union knots, with identical measurement vectors and saved-time buffer counts. No guard intervention or saturation is recorded. This supports the implemented calibration, integration order and constructor treatment of Proposition **??**; it is not a nonlinear-Fuzzy performance result or a general runtime-equivalence proof.

Separate previously completed reference calculations call the original 25-rule Fuzzy update directly, rather than advance the plant with a sector surrogate. Two trajectories use moderate and stronger gain coordinates, each with 2000 post-bootstrap updates. For the moderate configuration corresponding to Section **??**, the observed maximum fluid buffer deviation is approximately 17.314056818 frames, maximum $|X| + |Y|$ is 0.090130790, and maximum absolute correction is $3.21398377 \times 10^{-4}$. There are no observed stepwise contraction-envelope violations; frequencies remain in $[0.9998, 1.0002]$ including startup. These numerical checks are consistent with the reference certificate, but the all-time guarantee and strict PI comparison come from the interval-prefix and certified- tail argument of Theorem **??**, not from a finite floating-point trajectory. The reference marking is 1024; the delayed actual runs' marking is 1044.

5.5 Response to a post-centering latency burst

Table **??** reports the aligned moderate-gain comparison. Figure **??** shows the shared-time frequency, buffer and latency traces, including the clean counterfactuals. Fuzzy reduces RMS-amplitude IAE by 1.864% and mean command TV by 15.938%. Its disagreement RMS is 0.096% higher, instantaneous RMS peak is 2.396% higher, and target-referenced buffer excursion is one frame larger. These endpoints establish a tradeoff at this deterministic operating point, not a uniform disturbance-rejection advantage.

Both controllers remain within capacity with positive frequency and continuing logical phase progress, without recorded guard or saturation intervention. Including startup, the full-run maximum deviation is 24 frames for PI and 21 for Fuzzy. This is a favorable

Table 4: M2 metrics over $[2 \times 10^6, 4 \times 10^6)$ unless stated otherwise. Lower is preferable except for buffer headroom. The spread row is a supplementary descriptive calculation; no engineering spread threshold was predeclared for these runs.

Metric	Moderate PI	Full Fuzzy
Disagreement RMS R_W / ppm	9.67059	9.67988
Instantaneous RMS peak P_W / ppm	32.59447	33.37534
RMS-amplitude IAE $I_W/(10^6$ ppm · ticks)	9.94254	9.75722
Supplementary peak pairwise spread / ppm	69.14830	70.89161
Buffer excursion B_W^{obs} / frames	20	21
Minimum lower / upper buffer headroom / frames	1004 / 1017	1003 / 1017
Mean command TV / 10^{-4}	2.62204	2.20415
Largest command increment / 10^{-5}	4.10000	4.21763
Peak common-frequency difference from own clean / ppm	60.61864	61.26423
Component entry / confirmation after OFF / 10^6 ticks	0.8 / 1.1	0.8 / 1.1
Strict own-clean recovery by the horizon	Censored	Censored
Full-run buffer excursion, including startup / frames	24	21

Fuzzy endpoint for that full window, whereas the post-onset peaks are 20 and 21. Changing the window changes the ranking, so neither figure is used as a replacement for the other. Capacities are far from exhaustion.

Both first enter the window-qualified component recovery band at 2.9×10^6 and confirm it at 3.2×10^6. Thus the experiment does not show faster Fuzzy settling under this criterion. Strict own-clean recovery is censored for both. In the memory-diagnostic tail, the common-frequency changes relative to own clean are −59.40198 ppm for PI and −60.51069 ppm for Fuzzy. A stationary common frequency, frequency agreement between nodes, return to own clean, and return to frequency one are different properties.

5.6 Observed nonlinear mechanism and the first buffer peak

The M2 rule replay agrees with the full nonlinear implementation. All sampled Fuzzy input pairs remain in Ω_5, with disturbance-window maximum $|X| + |Y| = 2.001888$. The central identity of Proposition **??** therefore applies to those observed updates. However, this trajectory does not stay in the smaller Ω_1 input diamond used by the reference example. This is not evidence of a sector violation or instability; it is another reason, besides asynchrony, delay and quantization, that the example's invariant-region certificate is not a certificate for this actual burst.

The Section **??** decomposition into proportional, integral-like and accumulated nonlinear-residual terms reproduces the requested correction with maximum residual approximately 6.46×10^{-18}. In the diagnostic tail, the Fuzzy own-clean common-frequency shift decomposes into

$$-60.51069 = -57.61859 - 2.89210 \quad \text{ppm}, \tag{101}$$

where the two terms are the linear integral-like contribution and nonlinear memory; the proportional contribution is zero. PI's −59.40198 ppm shift is its integral contribution. This is exact same-trajectory accounting up to numerical arithmetic, not an independently evolving F_0 closed-loop ablation or a unique causal explanation of the physical common frequency.

Figure **??** also resolves the first buffer peak. PI and Fuzzy first attain their disturbance-window maximum target deficits at ON+39.105 and ON+40.731 ticks, whereas their first changed commands relative to own clean occur at ON+155.216 and ON+155.841. At the respective first extrema, clean/burst occupancies are 1024/1004 for PI and 1023/1003 for Fuzzy. Both same-time clean-minus-burst deficits are therefore 20 frames: target deficits of 20 and 21 combine that delivery deficit with clean-state offsets of zero and one. Consequently the first attainment is not a measurement of post-burst feedback effectiveness. The minimum is revisited later; we do not claim that every peak instance precedes feedback.

5.7 Independent paired random-latency validation

All 180 M3 validation runs are normal-admissible, with zero saved-state mismatches in complete-event replay. Table **??** summarizes the primary endpoint. The ratio estimand is

$$\widehat{r} = \frac{1}{30} \sum_{s=1}^{30} \frac{R_{W,s}^{\mathrm{Fuzzy}}}{R_{W,s}^{\mathrm{PI}}}, \tag{102}$$

where s indexes independent seeds. It is not the ratio of the two group means. Pointwise 95% confidence intervals (CIs) use 20,000 paired-seed percentile-bootstrap resamples with fixed resampling seeds. Time points, edges and bootstrap draws are not independent experimental replications. Regimes are analyzed separately; the intervals do not provide simultaneous multiple-endpoint coverage. There are no excluded pairs in the reported comparisons.

Relative to moderate PI, Fuzzy has higher RMS in all 30 seeds of each regime: mean paired increases are 19.245% and 6.791%. The lower RMS relative to stronger fixed PI is comparator-specific; moderate PI already has lower RMS than Fuzzy, so that comparison does not establish an inherent advantage of fuzzy inference over suitable PI tuning. The frozen random-latency hypothesis of lower disagreement RMS is therefore not supported against the primary baseline in these regimes.

Within the disturbance windows, all runs have at most seven frames of actual target deviation and at least 1017 frames of lower/upper headroom. These extrema exclude startup. They establish finite-run

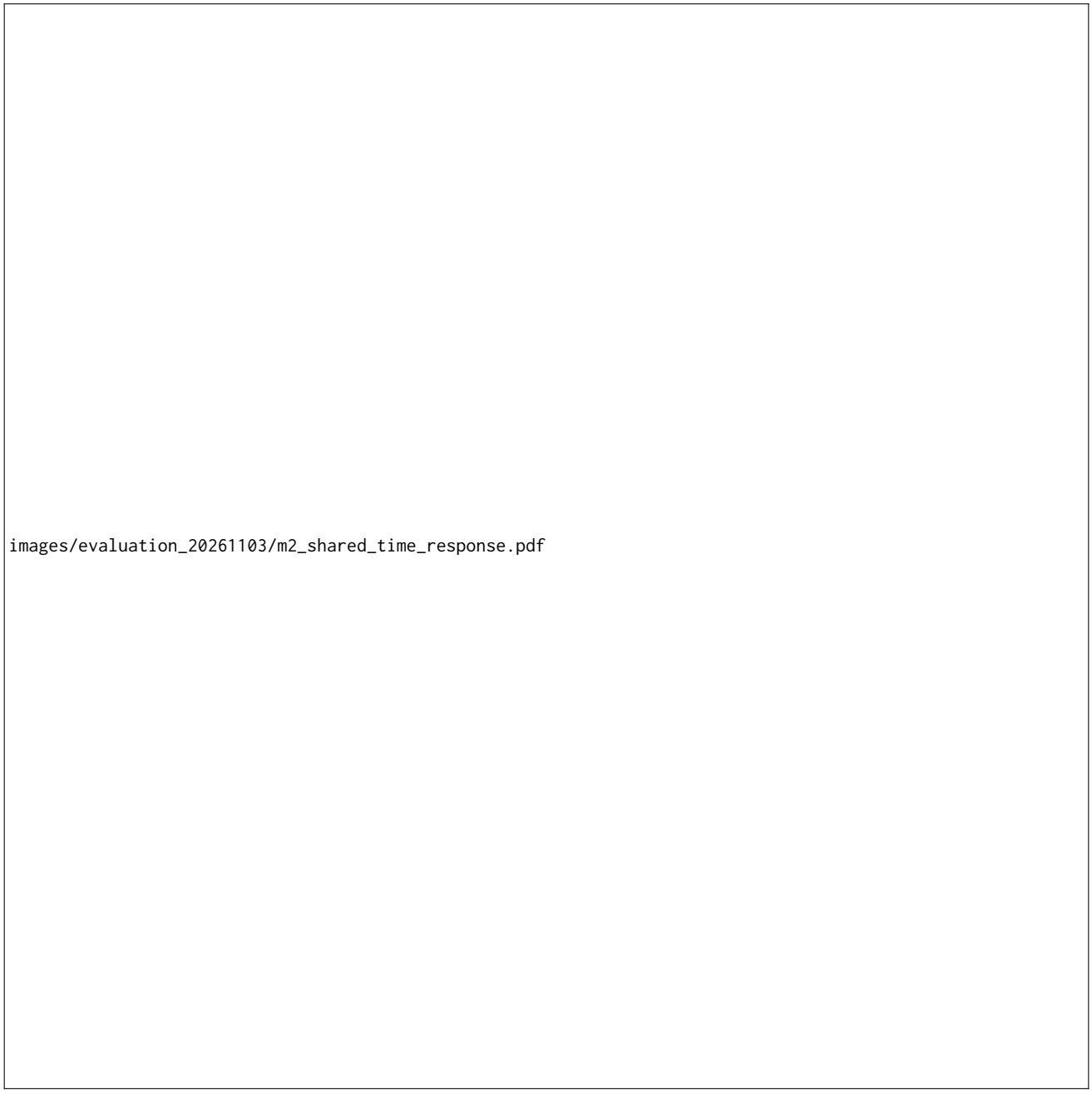


Figure 1: Aligned post-centering burst experiment. Continuous-phase Callisto uses the fixed moderate PI coordinates and their full-Fuzzy calibration. Both controllers have their own clean counterfactual. Desired latency on one directed channel changes from 20 to 40 during $[2 \times 10^6, 2.1 \times 10^6)$. Frequency histories use held corrections; buffer overview samples are for display, while reported buffer statistics come from the complete reconstructed event stream.

admissibility in this interior operating region, not a larger Fuzzy safe set. The random result also does not extend the M2 TV reduction: mean paired Fuzzy/moderate TV ratios are 1.03045 and 1.03153. Figure **??** retains buffer, effort and common-frequency diagnostics. Mean tail common-frequency shifts from each controller's pre-disturbance level are approximately −156.682 versus −72.576 ppm at $T_c = 1000$, and −76.540 versus −67.816 ppm at $T_c = 10000$ (Fuzzy versus moderate PI). These finite-horizon shifts and trends are reported separately from relative synchronization; no absolute-frequency loop is present.

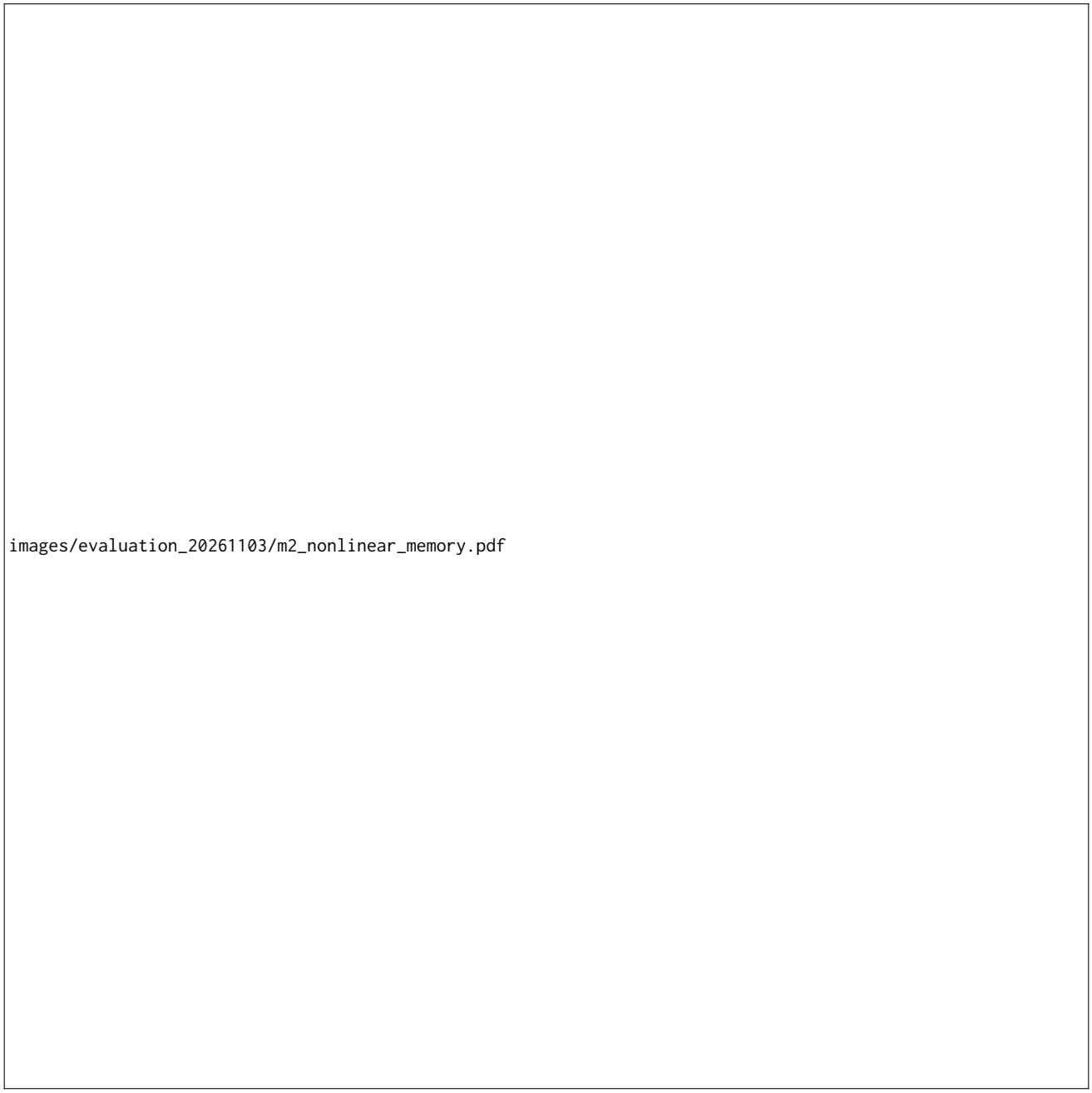


Figure 2: Nonlinear mechanism diagnostics reusing the four M2 runs. The panels show the exact central residual $F - F_0$, common-frequency memory accounting, command activity and complete early buffer events. The hatched F_0 activity bar is a replay on Fuzzy's measured inputs, not a third closed-loop experiment. The residual surface is multiplied by negative K_u in the controller. The memory panel uses $[3.7 \times 10^6, 4 \times 10^6)$; activity uses the main disturbance window. First peak attainment precedes the first changed command, but the peak can be revisited after feedback begins.

5.8 What the completed evaluation establishes

The evidence supports the implementation-consistent PI calibration, the exact nonlinear mechanism on observed M2 inputs, and the conditional reference-model buffer-headroom theorem. Actual Callisto demonstrates endpoint-dependent response shaping: a smaller full-run buffer excursion and lower post-burst IAE/TV at the stated M2 point, alongside higher post-onset peaks and no faster component

Table 5: M3 primary disturbance-window disagreement RMS. Controller entries are means over 30 seeds in ppm. Ratios are means of paired seed ratios with pointwise 95% CIs. A ratio below one favors Fuzzy.

Profile grid	PI moderate	PI stronger	Fuzzy	Fuzzy/moderate [CI]	Fuzzy/stronger [CI]
$T_c = 1000$	4.911	9.347	5.856	1.1925 [1.1722,1.2150]	0.6266 [0.6155,0.6389]
$T_c = 10000$	6.696	9.699	7.152	1.0679 [1.0624,1.0736]	0.7365 [0.7184,0.7554]

recovery. The independent M3 study limits any general random-jitter advantage claim for the frozen controller. These outcomes are consistent with the paper's conditional mechanism-and-tradeoff formulation; they do not support uniform Fuzzy superiority.

The completed actual campaigns do not test the boundary of the reference invariant set. Their large buffer reserves cannot substitute for certified-safe, uncertified-but-observed-safe and failed cases around a common boundary. Nor do finite observations prove infinite-horizon safety or the reference theorem's optional frequency-spread tolerance in asynchronous delayed execution. Observed phase progress and capacity compliance support normal model execution; application-payload correctness and a formal refinement proof of logical synchrony are outside this evaluation.

Finally, these are fixed-controller comparisons, not a completed equal-budget tuning/held-out-test study. A separate preregistered tuning campaign has not completed its audit, selection and untouched tests; its partial results are excluded from the present performance tables. The evaluation also provides no P/GABA comparison, switched FFP or observer result, dropout recovery, serialized-link hardware test, or meaningful controller runtime/energy ranking. Consequently, practical adoption beyond the reported operating points remains an empirical question even though the reference-model headroom benefit is mathematically established.

6 Discussion and Limitations

Draft status: to be written. The full discussion remains pending. The certificate in Section **??** applies to its specified synchronous, zero-delay fluid reference and prescribed initialization. It does not certify asynchronous, delayed, quantized Callisto execution or establish universal superiority over PI.

7 Conclusion

Draft status: to be written. Complete this section after the evaluation and discussion are integrated, keeping reference-model guarantees distinct from measured Callisto results.

Acknowledgments

Draft status. Funding and acknowledgments are pending. Remove identifying material when preparing an anonymous submission.

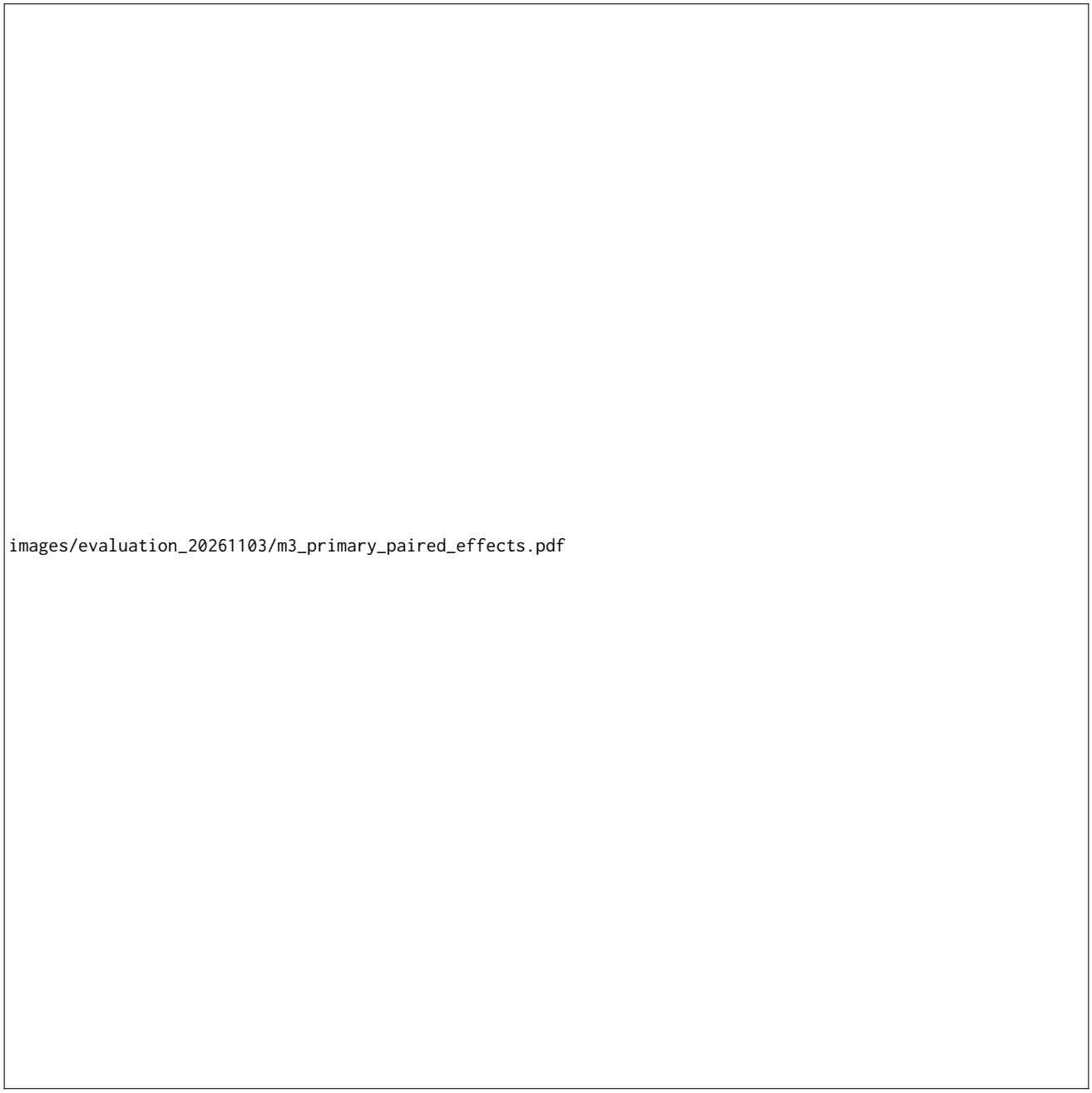


Figure 3: Independent paired random-latency validation: primary disagreement RMS and paired Fuzzy/PI effects for two physical profile grids. Thirty seeds per regime are shared across the three fixed controllers. The moderate comparator is the primary axis-matched PI baseline; stronger PI is a separate fixed candidate. Ratios and pointwise confidence intervals follow the paired-seed protocol, rather than resampling correlated controller updates.

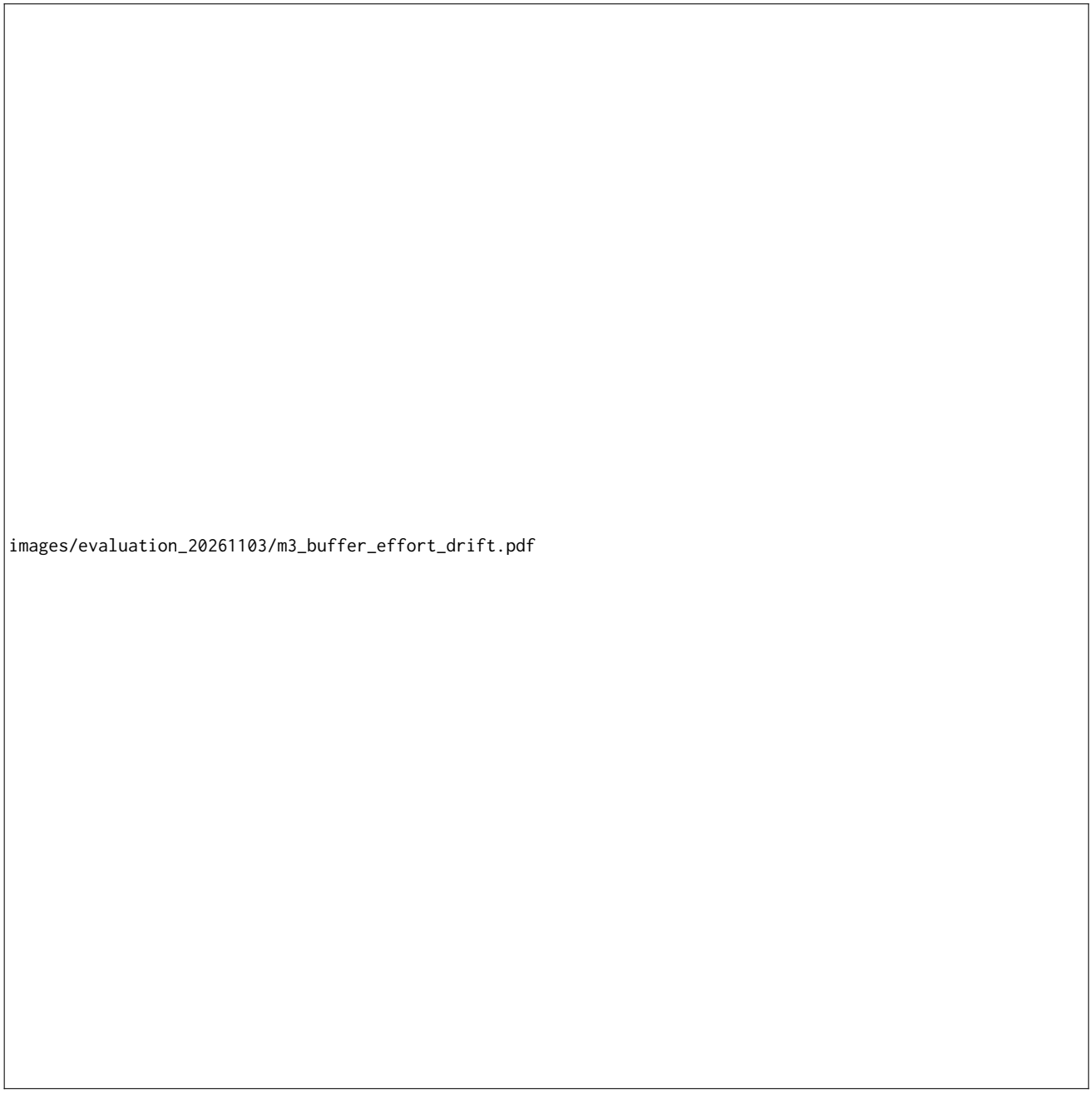


Figure 4: Secondary M3 diagnostics over the declared windows: complete-event buffer excursions, command activity and common-frequency shift. The same 30 paired seeds underlie each regime; additional endpoints do not create additional independent replications. Buffer extrema refer to the disturbance window, not the startup trajectory.